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Operating mechanism of circuit breaker, and circuit breaker

Abstract

An operating mechanism of a circuit breaker, relating to the field of low-voltage electric appliances. The operating mechanism is provided with a double-slider mechanism, a first slider mechanism comprising a first crank, a first connecting rod, a sliding rail and a slider, and a second slider mechanism comprising a contact support, a second connecting rod, a sliding rail and a slider. The double-slider mechanism reduces the space requirement of the operating mechanism while increasing the opening distance of a moving contact, and the breaking performance is good. A circuit breaker comprising the operating mechanism, which provides good breaking performance.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) The present application is a 35 U.S.C. §§ 371 national stage application of International Application No. PCT/CN2022/084552, filed Mar. 31, 2022, which claims priority to Chinese Patent Application No. CN202110355214.0, filed Apr. 1, 2021, the contents of which are incorporated herein by reference.

TECHNICAL FIELD

(2) The present invention relates to the field of low-voltage electrical appliances, and more particularly to an operating mechanism of a circuit breaker and the circuit breaker including the operating mechanism.

BACKGROUND ART

(3) With the continuous improvement of an electric energy system, the voltage requirements for a circuit breaker are getting higher and higher. In order to meet the needs of the circuit breaker for a breaking high voltage, increasing an opening distance between a moving contact and a static contact is a common design method, which generally includes the following two design schemes.

(4) One scheme is to increase a length of a moving contact arm, but has the following disadvantages: in order to ensure the reliable connection performance between the moving contact and the static contact, there must be sufficient pressure between the moving contact and the static contact; with the lengthening of the moving contact arm, a required contact spring force will increase sharply, and the force of a spring of the operating mechanism will also increase sharply; and meanwhile, the service life of the circuit breaker is shortened due to the increase in operating force.

(5) The other scheme is to adjust a four-link structure of the existing operating mechanism to increase the opening distance. Due to the high correlation between movements of respective connecting rods in the four-link structure, if a contact support rotates at a larger angle while driving the moving contact to be disconnected, a jump buckle and an upper connecting rod also need to rotate at a larger angle. However, the rotation angle of the jump buckle is directly related to a latch and a handle, such that the handle and the latch need to rotate at a larger angle, which puts forward higher requirements for space. In addition, the length, position, and elasticity of a spring adapted to the jump buckle change, and accordingly, the larger angle of the jump buckle in rotation will slow down a movement speed of the entire mechanism.

SUMMARY OF THE INVENTION

(6) An object of the present invention is to overcome the defects of the prior art and improve an operating mechanism of a circuit breaker, so that an opening distance of a moving contact is increased without increasing a space requirement of the operating mechanism, and a good breaking performance is achieved.

(7) The present invention further provides a circuit breaker including the operating mechanism, and the circuit breaker has a good breaking performance.

(8) In order to achieve the above object, the present invention adopts the following technical solutions:

(9) The present invention provides an operating mechanism of a circuit breaker, comprising a bracket; a rocker arm assembly and a jump buckle pivotally disposed on the bracket respectively, as well as a first crank, a first spring, a first connecting rod and a contact support, wherein one end of the first crank is pivotally disposed on the jump buckle around a first axis, and the other end of the first crank is rotatably connected to one end of the first connecting rod; one end of the first spring is

connected to the rocker arm assembly, and the other end of the first spring is connected to the rotating connection position of the first crank and the first connecting rod; and the operating mechanism further comprises slide rails, a slider and a second connecting rod; the slider is slidably disposed on the slide rails and is rotatably connected to the other end of the first connecting rod; one end of the second connecting rod is rotatably connected to the slider, and the other end of the second connecting rod is rotatably connected to the contact support; and the contact support is driven to rotate around a third axis.

(10) Further, the operating mechanism further comprises an auxiliary limiting structure, one end of the auxiliary limiting structure is rotatably connected to the second connecting rod, and the other end of the auxiliary limiting structure is rotatably connected to the bracket or a housing of the circuit breaker.

(11) Further, the auxiliary limiting structure is a third crank, one end of the third crank is rotatably connected to the second connecting rod, and the other end of the third crank is rotatably connected to the bracket or the housing of the circuit breaker.

(12) Further, the slide rails are disposed on the bracket or the housing of the circuit breaker, and the slide rails are of a groove-like structure or a hole-like structure.

(13) Further, each slide rail is straight, arc-shaped, triangular, or a combined shape of straight line and arc.

(14) Further, the bracket comprises two bracket arms disposed to face each other at intervals, and each bracket arm is provided with a slide rail, and two ends of the slider are respectively slidably disposed on the two slide rails.

(15) Further, the slide rails is slide holes, and the slider is a sliding shaft, two ends of which are respectively disposed in the two slide holes.

(16) Further, the second connecting rod has an arc or straight plate structure

(17) Further, the other end of the second connecting rod is rotatably connected to the contact support through the first connecting shaft;

(18) Further, the bracket comprises a second avoidance hole for the first connecting shaft to pass through for avoiding the first connecting shaft.

(19) Further, the operating mechanism includes a plurality of contact supports arranged side by side at intervals, and the respective contact supports are linked by means of a linkage shaft; and the housing of the circuit breaker is provided with a first avoidance hole for the linkage shaft to pass through for avoiding the linkage shaft.

(20) Further, the operating mechanism further includes a latch and a re-buckle disposed on the bracket respectively, wherein a jump buckle is in latching fit with the latch, and the latch is in limiting fit with the re-buckle.

(21) The present invention provides another operating mechanism of a circuit breaker, comprising a bracket; a rocker arm assembly and a jump buckle pivotally disposed on the bracket respectively, as well as a first crank, a first spring, a first connecting rod and a contact support, wherein one end of the first crank is pivotally disposed on the jump buckle around a first axis, and the other end of the first crank is rotatably connected to one end of the first connecting rod; one end of the first spring is connected to the rocker arm assembly, and the other end of the first spring is connected to the rotating connection position of the first crank and the first connecting rod; and the operating mechanism further comprises slide rails, a slider, a second crank and a second connecting rod; the slider is slidably disposed on the slide rails and is rotatably connected to the other end of the first connecting rod; the second crank comprises a second crank supporting portion, a second crank connecting portion, and a second crank driving portion, wherein the second crank is pivotally disposed through the second crank supporting portion, and the second crank is connected to the contact support through the second crank driving portion; one end of the second connecting rod is rotatably connected to the slider, and the other end of the second connecting rod is rotatably connected to the second crank connecting portion; and the contact support is driven to rotate

around a third axis.

(22) Further, the second crank is pivotally disposed on the bracket or the housing of the circuit breaker through the second crank supporting portion; the second crank supporting portion and the second crank driving portion are respectively disposed on two ends of the second crank; and the second crank connecting portion is disposed between the second crank supporting portion and the second crank driving portion.

(23) Further, the second crank driving portion is rotatably connected to the contact support through a linkage shaft.

(24) Further, the housing of the circuit breaker is provided with a first avoidance hole for the linkage shaft to pass through for avoiding the linkage shaft; the other end of the second connecting rod is rotatably connected to the second crank connecting portion through the first connecting shaft; and the bracket is provided with a second avoidance hole for the first connecting shaft to pass through for avoiding the first connecting shaft.

(25) Further, the center of rotation of the second crank is a ninth axis, and the ninth axis is parallel to or coincided with the third axis.

(26) Further, the slide rails are disposed on the bracket or the housing of the circuit breaker, and the slide rails are of a groove-like structure or a hole-like structure.

(27) Further, each slide rail is straight, arc-shaped, triangular, or a combined shape of straight line and arc.

(28) Further, the second connecting rod has an arc or straight plate structure

(29) Further, the operating mechanism further includes a latch and a re-buckle disposed on the bracket respectively, wherein a jump buckle is in latching fit with the latch, and the latch is in limiting fit with the re-buckle.

(30) Further, the operating mechanism further comprises a moving contact, the moving contact is a single-breakpoint contact or more breakpoints contact.

(31) Further, the operating mechanism further comprises a latch and a re-buckle which are pivotally disposed on the bracket respectively; the jump buckle is in latching fit with the latch, and the latch is in limiting fit with the re-buckle; the rocker arm assembly comprises a handle, a rocker arm fixedly connected to the handle, and a reset structure for driving the jump buckle to rotate to re-buckle with the latch, the rocker arm being pivotally disposed on the bracket; the first crank comprises a crank limiting portion, which is in limiting fit with the jump buckle; two ends of a swing stroke of the rocker arm are a first stroke end and a second stroke end, respectively; two ends of the first spring are a first spring end and a second spring end, respectively, which are respectively connected to the rocker arm assembly and the first crank.

(32) Further, in the case that the operating mechanism is in a closed state, the rocker arm swings toward the second stroke end and drives the first spring end to rotate around the second spring end, until the first spring passes a first dead center position; the first spring drives the first crank to rotate in a second direction and drives the rocker arm to swing to the second stroke end; the first crank drives the slider through the first connecting rod to slide along the slide rails; and the slider drives the contact support through the second connecting rod to rotate in a first direction to a breaking position, so that the operating mechanism is switched to a disconnected state;

(33) in the case that the operating mechanism is in the disconnected state, the rocker arm swings toward the first stroke end and drives the first spring end to rotate around the second spring end, until the first spring passes the first dead center position; the first spring drives the first crank to rotate in the first direction, such that the crank limiting portion is in limiting fit with the jump buckle, thereby preventing the first crank from rotating in the first direction; meanwhile, the first spring drives the rocker arm to swing to the first stroke end, and the first crank drives the slider through the first connecting rod to slide along the slide rails; and the slider drives the contact support through the second connecting rod to rotate in the first direction to a closing position, so that the operating mechanism is switched to a closed state; the first direction and the second

direction are opposite to each other;

(34) Further, in the case that the operating mechanism is in the closed state, the re-buckle rotates to release the limiting fit from the latch, and the latch rotates to release the latching fit from the the jump buckle; the jump buckle rotates and drives the first crank to rotate synchronously, the first crank drives the slider through the first connecting rod to slide along the slide rails, and meanwhile, the slider drives the contact support through the second connecting rod to rotate in the second direction to the breaking position; the first spring drives the rocker arm to swing toward the second stroke end to the reset structure till being in limiting fit with the jump buckle, and the operating mechanism is switched to a tripped state; and in the case that the operating mechanism is in the tripped state, the rocker arm swings to the second stroke end, and the rocker arm drives the jump buckle through the reset structure to rotate to be in latching fit with the latch; and meanwhile, the latch rotates to be in limiting fit with the re-buckle, and the operating mechanism is switched to the disconnected state.

(35) The present invention further provides a circuit breaker, comprising the operating mechanism of the circuit breaker.

(36) According to an operating mechanism of a circuit breaker, a first crank, a first connecting rod, slide rails and a slider form a first slider mechanism, and a contact support, a second connecting rod, the slide rails and the slider form a second slider mechanism. The dual-slider mechanism reduces the correlation between a rotation range of the contact support and rotation angles of the jump buckle and the first connecting rod, thereby achieving the purpose of increasing the opening distance of the moving contact and reducing the space requirement of the operating mechanism. Moreover, the reliable operating performance of the operating mechanism is ensured. In addition, a third crank assists in limiting the operation of the second connecting rod, and the synergy of the second connecting rod and the third crank makes the movement of the second connecting rod more precise, which is beneficial to improve the working reliability and stability of the operating mechanism.

(37) According to another operating mechanism of the circuit breaker, a first crank, a first connecting rod, slide rails and a slider form a first slider mechanism, and a contact support, a second connecting rod, a second crank, the slide rail and the slider form a second slider mechanism. The dual-slider mechanism reduces the correlation between a rotation range of the contact support and rotation angles of the jump buckle and the first connecting rod, thereby achieving the purpose of increasing the opening distance of the moving contact and reducing the space requirement of the operating mechanism. Moreover, the reliable operating performance of the operating mechanism is ensured. Moreover, the second crank is beneficial to improve the assembling accuracy of the operating mechanism and reduce the assembling error.

(38) The circuit breaker of the present invention includes the operating mechanism, which has a compact structure and good breaking performance.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a principle schematic diagram of an operating mechanism of the present invention, in which the operating mechanism is in a closed state;

(2) FIG. 2 is a principle schematic diagram of the operating mechanism of the present invention, in which the operating mechanism is in a disconnected state;

(3) FIG. 3 is a principle schematic diagram of the operating mechanism of the present invention, in which the operating mechanism is in a tripped state;

(4) FIG. 4 is a schematic structural diagram of the operating mechanism of the present invention, in which the operating mechanism is in a closed state;

- (5) FIG. 5 is a schematic structural diagram of the operating mechanism of the present invention, in which the operating mechanism is in a disconnected state;
- (6) FIG. 6 is a schematic structural diagram of the operating mechanism of the present invention, in which the operating mechanism is in a tripped state;
- (7) FIG. 7 is a principle schematic diagram of a circuit breaker of the present invention, in which the circuit breaker is in a closed state;
- (8) FIG. 8 is a principle schematic diagram of the circuit breaker of the present invention, in which the circuit breaker is in a disconnected state;
- (9) FIG. 9 is a principle schematic diagram of the circuit breaker of the present invention, in which the circuit breaker is in a tripped state;
- (10) FIG. 10 is a principle schematic diagram of the circuit breaker of the present invention, in which a moving contact point is respectively disposed on two ends of a moving contact and is respectively matched with two static contacts, and the circuit breaker is in a closed state;
- (11) FIG. 11 is a schematic structural diagram of the circuit breaker of the present invention, in which the circuit breaker is in a closed state;
- (12) FIG. 12 is a schematic structural diagram of the circuit breaker of the present invention, in which the circuit breaker is in a disconnected state;
- (13) FIG. 13 is a schematic structural diagram of the circuit breaker of the present invention, in which the circuit breaker is in a tripped state;
- (14) FIG. 14 is a schematic structural diagram of the circuit breaker of the present invention, in which the moving contacts are repulsed by an electric repulsion force;
- (15) FIG. 15 is a schematic structural diagram of the circuit breaker of the present invention, in which a contact spring rotates to a second dead center position;
- (16) FIG. 16 is a schematic structural diagram of the circuit breaker of the present invention, in which the contact spring locks a moving contact;
- (17) FIG. 17 is a schematic structural diagram of the operating mechanism of the present invention, in which a jump buckle and a bracket are fixed together by a positioning shaft pin, a second spring shaft is in limiting fit with the jump buckle, a first crank is in limiting fit with the jump buckle, and a spacing between the second spring shaft and a first spring shaft is less than or equal to the length of the first spring;
- (18) FIG. 18 is a schematic structural diagram of the operating mechanism of the present invention, in which a first spring is mounted on the first spring shaft and the second spring shaft compared to FIG. 17;
- (19) FIG. 19 is a schematic structural diagram of the operating mechanism of the present invention, in which a positional relationship between the first spring and the jump buckle is shown;
- (20) FIG. 20 is a schematic structural diagram of the operating mechanism of the present invention, in which a reset structure is mounted on a rocker arm compared to FIG. 18;
- (21) FIG. 21 is a schematic structural diagram of the operating mechanism of the present invention, in which a positioning shaft pin is detached compared to FIG. 20;
- (22) FIG. 22 is a schematic projection view of a bracket of the present invention, in which at least a V-shaped groove is shown;
- (23) FIG. 23 is a stereoscopic structural diagram of the bracket of the present invention;
- (24) FIG. 24 is a schematic projection view of the bracket of the present invention, in which at least a positional relationship and spacing between two jump buckle positioning arms are shown;
- (25) FIG. 25 is a schematic diagram of the connection of the jump buckle and the first crank of the present invention;
- (26) FIG. 26 is a schematic diagram of an assembly structure of the bracket, the jump buckle and a jump buckle shaft of the present invention;
- (27) FIG. 27 is a schematic structural diagram of a rocker arm assembly of the present invention;
- (28) FIG. 28 is a schematic structural diagram from another perspective of the rocker arm assembly

of the present invention;

(29) FIG. **29** is a schematic structural diagram of a moving contact assembly of the present invention;

(30) FIG. **30** is a schematic structural diagram of the moving contact assembly of the present invention, in which at least a connection relationship among a moving conductive rod, a first clamping arm and a second clamping arm is shown;

(31) FIG. **31** is a schematic structural diagram of an implementation of a conductor of the present invention;

(32) FIG. **32** is a schematic structural diagram of a fastener of the present invention;

(33) FIG. **33** is a schematic structural diagram of a first embodiment of a moving contact of the present invention;

(34) FIG. **34** is a schematic structural diagram of another implementation of a conductor of the present invention;

(35) FIG. **35** is a schematic structural diagram of an implementation of the moving contact mechanism of the present invention;

(36) FIG. **36** is a schematic structural diagram of a contact support of the present invention;

(37) FIG. **37** is a schematic structural diagram of a moving contact insulator of the present invention;

(38) FIG. **38** is a schematic diagram of an assembly structure of the moving contact assembly, the static contact and a unit housing of the present invention;

(39) FIG. **39** is a schematic structural diagram of a circuit breaker pole of the present invention, in which an assembly relationship among the moving contact mechanism, a first push rod and a second push rod is shown;

(40) FIG. **40** is a schematic structural diagram of the circuit breaker pole of the present invention, in which a cooperative relationship among the moving contact, the first push rod, the second push rod and the unit housing is shown;

(41) FIG. **41** is a schematic diagram of an assembly structure of the operating mechanism, a quick tripping device and the circuit breaker pole of the present invention;

(42) FIG. **42** is a schematic diagram of an assembly structure of a first intermediate push rod and a first intermediate shaft of the present invention;

(43) FIG. **43** is a schematic structural diagram of the circuit breaker of the present invention, in which an assembly relationship between the operating mechanism and respective circuit breaker poles is shown;

(44) FIG. **44** is a schematic structural diagram of the circuit breaker pole of the present invention, in which a cooperative relationship between the quick tripping device of each circuit breaker pole and the operating mechanism is shown;

(45) FIG. **45** is a schematic structural diagram of another implementation of the moving contact mechanism of the present invention;

(46) FIG. **46** is a schematic structural diagram of a second embodiment of a moving contact of the present invention; and

(47) FIG. **47** is a schematic structural diagram of the operating mechanism of the present invention, in which one end of the first crank is pivotally disposed on the bracket.

(48) FIG. **48** is another schematic structural diagram of the circuit breaker of the present invention, in which the circuit breaker is in a closed state;

(49) FIG. **49** is another schematic structural diagram of the circuit breaker of the present invention, in which the circuit breaker is in a disconnected state.

DETAILED DESCRIPTIONS OF THE PREFERRED EMBODIMENTS

(50) The specific implementation of the circuit breaker of the present invention will be further described below with reference to the embodiments given in FIGS. **1** to **47**. The circuit breaker of the present invention is not limited to the description of the following embodiments.

(51) As shown in FIGS. 7-16, 38-41 and 43, the circuit breaker of the present invention includes an operating mechanism **100** and at least one circuit breaker pole **300**; the operating mechanism **100** includes at least one group of moving contact mechanisms. Each moving contact mechanism includes a contact support **110** pivotally arranged around a third axis **111s** and a moving contact **9** disposed on the contact support **110** to rotate synchronously with the contact support **110**; each circuit breaker pole **300** includes a static contact **18** which is in one-to-one cooperation with the moving contact mechanism to form a contact system; at least one group of contact systems is disposed in each circuit breaker pole **300**; and the operating mechanism **100** actuates, so that the moving contact **9** and the static contact **18** are disconnected or closed, to implement a closing or disconnecting operation of the circuit breaker.

(52) Further, the moving contact **9** rotates around a contact axis, and the contact axis coincides with the third axis **111s**, or the contact axis is parallel to (but not coincident with) the third axis **111s**. It should be pointed out that the circuit breaker of the present invention further includes a moving contact rotating shaft for supporting the rotation of the moving contact **9**. An axis of the moving contact rotating shaft coincides with the third axis **111s**, and the moving contact **9** is provided with a moving contact shaft hole **901** matched with the moving contact rotating shaft. In practical applications, a diameter of the moving contact shaft hole **901** is slightly larger than that of the moving contact rotating shaft, so as to ensure the rotation flexibility of the moving contact **9**. In the case that the circuit breaker of the present invention is driven by the operating mechanism **100** to be disconnected or closed normally, the moving contact **9** acts synchronously with the contact support **110**, and rotation axis lines of the moving contact **9** and the contact support **110** coincide at this time. In the case that the moving contact **9** is repelled to rotate relative to the contact support **110** in response to the occurrence of a short-circuit fault in the circuit breaker of the present invention, the inside surface of the movable contact shaft hole **901** and the moving contact rotating shaft support the moving contact **9** to rotate in a tangential manner. At this time, the rotation axis line of the moving contact **9** is parallel to the third axis **111s**.

(53) Further, as shown in FIG. 43, the circuit breaker of the present invention includes a plurality of circuit breaker poles **300** arranged side by side, and the moving contact mechanisms in the respective circuit breaker poles **300** are linked.

(54) Further, as shown in FIGS. 7-9, the movable contact **9** is a single-breakpoint contact, wherein a moving contact point is disposed at one end of the moving contact **9**, and the other end of the moving contact **9** is in driving fit with the contact support **110**; or, as shown in FIG. 10, the moving contact **9** is a double-breakpoint contact, wherein a moving contact point is respectively disposed at two ends of the moving contact **9**, and the middle part of the moving contact **9** is in driving fit with the contact support **110**. It should be pointed out that the moving contact **9** may also include more breakpoints.

(55) As shown in FIG. 43, the following is an implementation of a housing of the circuit breaker of the present invention: the circuit breaker of the present invention further includes a circuit breaker housing **3**, wherein the operating mechanism **100** and the respective circuit breaker poles **300** arranged side by side are disposed inside the circuit breaker housing **3** respectively; each circuit breaker pole **300** further includes a unit housing **120**, and the contact system of each circuit breaker pole **300** is disposed in the corresponding unit housing **120**; and the circuit breaker housing includes a circuit breaker shell **3** and the unit housing **120**.

(56) The following is another implementation of the housing of the circuit breaker of the present invention: the circuit breaker of the present invention further includes a circuit breaker shell **3**, wherein the respective circuit breaker poles **300** are arranged side by side inside the circuit breaker shell **3** at intervals, and a phase separation partition is arranged between adjacent circuit breaker poles **300** to ensure a sufficient electric clearance and creepage distance between the circuit breaker poles **300**; and the circuit breaker housing includes the circuit breaker shell **3** and the phase separation partition.

(57) The circuit breaker of the present invention will be further described below in conjunction with the accompanying drawings and specific embodiments.

(58) As shown in FIGS. **1-16, 43** and **44**, the circuit breaker in this embodiment, preferably a molded case circuit breaker, includes an operating mechanism **100** and a plurality of circuit breaker poles **300**; the operating mechanism **100** includes a moving contact mechanism. The moving contact mechanism includes a contact support **110** pivotally arranged around a third axis **111s** and a moving contact **9** disposed on the contact support **110** to rotate synchronously with the contact support **110**; each circuit breaker pole **300** includes a static contact **18** which is in one-to-one cooperation with the moving contact mechanism to form a contact system; at least one group of contact systems is disposed in each circuit breaker pole **300**; and the operating mechanism **100** actuates, so that the moving contact **9** and the static contact **18** are disconnected or closed, to implement a closing or disconnecting operation of the circuit breaker.

(59) Specifically, as shown in FIGS. **43** and **44**, the circuit breaker in this embodiment is a three-phase circuit breaker, including three circuit breaker poles **300** (respectively used for connecting or breaking a three-phase circuit of a power supply) arranged side by side; the operating mechanism **100** is mounted on the circuit breaker pole **300** located in the middle; and moving contact mechanisms of the three circuit breaker poles **300** are linked (as shown in FIG. **3**, the moving contact mechanisms of the three circuit breaker poles **300** are further linked by a linkage shaft **5**). Of course, the number of the circuit breaker poles **300** can be adjusted according to actual needs. For example, the number of the circuit breaker poles **300** may be 2, which are matched with a two-phase power supply; or, the number of the circuit breaker poles **300** may also be 4, which are used for a three-phase four-wire circuit; or, the number of the circuit breaker poles **300** may also be 1, which is matched with a single-phase circuit.

(60) As shown in FIG. **43**, the following is a first embodiment of the housing of the circuit breaker in this embodiment: the circuit breaker in this embodiment further includes a circuit breaker shell **3**, wherein the operating mechanism **100** and the respective circuit breaker poles **300** arranged side by side are disposed in the circuit breaker shell **3** respectively; each circuit breaker pole **300** further includes a unit housing **120**, and the contact system of each circuit breaker pole **300** is disposed in the corresponding unit housing **120**; the contact support **110** is pivotally disposed on the unit housing **120**; and the circuit breaker housing includes the circuit breaker shell **3** and the unit housing **120**.

(61) Further, as shown in FIG. **38**, the contact support **110** includes support shaft grooves **111** respectively formed in both sides of the contact support **110**, and the unit housing **120** includes housing shaft columns **120-2** matched with the support shaft grooves **111**. Further, as shown in FIGS. **38** and **41**, the unit housing **120** includes two half housings that are matched with each other, and the two housing shaft columns **120-2** are respectively disposed on the inner walls of the two half housings.

(62) The circuit breaker in this embodiment can also achieve the following technical effects: the operating mechanism **100** (except the moving contact mechanism) is integrally disposed outside the unit housing **120**, and the contact system is disposed inside the unit housing **120**, which, firstly, prevents electric arc particles generated by the closing/breaking of the moving contact from being deposited on the operating mechanism **100** to affect the operating performance of the operating mechanism **100**; secondly, improves the insulation performance of the circuit breaker to ensure the personal safety of users; and thirdly, is beneficial to realize the modularized assembly of the operating mechanism **100**.

(63) The following is a second embodiment of the housing of the circuit breaker in this embodiment: the circuit breaker in this embodiment further includes a circuit breaker shell **3**, wherein the respective circuit breaker poles **300** are arranged side by side inside the circuit breaker shell **3** at intervals, and a phase separation partition is disposed between adjacent circuit breaker poles **300**; and the circuit breaker housing includes the circuit breaker shell **3** and the phase

separation partition.

(64) The first embodiment of the operating mechanism **100** is shown in FIGS. 7-13. The operating mechanism **100** is provided with a dual-slider mechanism, which can reduce the correlation between rotation angles of the operating mechanism and the contact system, and increase an opening distance of the moving contact, without increasing a space requirement of the operating mechanism, to be specific: as shown in FIGS. 7-13, the operating mechanism **100** includes a bracket **50**, a rocker arm assembly and a jump buckle **60** pivotally disposed on the bracket **50** respectively, as well as a first crank **30**, a first spring **22**, a first connecting rod **27** and a contact support **110**, wherein one end of the first crank **30** is pivotally disposed on the jump buckle **60** around a first axis **67m**, and the other end of the first crank **30** is rotatably connected to one end of the first connecting rod **27**; one end of the first spring **22** is connected to the rocker arm assembly, and the other end of the first spring **22** is connected to the rotating connection position of the first crank **30** and the first connecting rod **27**; the operating mechanism **100** further includes slide rails **25**, a slider **26** and a second connecting rod **29**; the slide rails **25** are disposed on the bracket **50** or the circuit breaker housing, and the slider **26** is slidably disposed on the slide rails **25** and is rotatably connected to the other end of the first connecting rod **27**; and one end of the second connecting rod **29** is rotatably connected to the slider **26**, and the other end of the second connecting rod **29** is rotatably connected to the contact support **110**, and the contact support **110** is driven to rotate around the third axis **111s**. According to the operating mechanism **100**, the first crank **30**, the first connecting rod **27**, the slide rails **25** and the slider **26** form a first slider mechanism, and the contact support **110**, the second connecting rod **29**, the slide rails **25** and the slider **26** form a second slider mechanism. The first slider mechanism cooperates with the second slider mechanism to reduce the correlation of rotation ranges of the jump buckle **60**, the first connecting rod **27** and the contact support **110**, thereby increasing the opening distance of the moving contact, without increasing the space requirement of the operating mechanism. Moreover, the reliable operating performance of the operating mechanism is ensured.

(65) Further, the slide rails **25** are of a groove-like structure or a hole-like structure. Further, in the case that the slide rail **25** is of the groove-like structure, the slide rail **25** may be disposed on the inside wall of the bracket **50** or the inside wall of the unit housing **120** (a phase separation partition is disposed between adjacent circuit breaker poles **300** in the case that the circuit breaker pole **300** is not provided with a unit housing **120**, and the slide rail **25** is disposed on a sidewall of the phase separation partition), and the slide rail **25** does not penetrate through the bracket **50** or the unit housing **120** (or the phase separation partition) in a thickness or depth direction. Further, two ends of the slide rail **26** are respectively disposed in the two slide rails **25**, and one end of the first connecting rod **27** connected to the slider **26** and one end of the second connecting rod **29** connected to the slider **26** are both disposed in a space between the two slide rails **25**. In the case that the slide rail **25** is of the hole-like structure, the slide rail **25** may be disposed on the inside wall (as shown in FIGS. 4-6) of the bracket **50** or the inside wall of the unit housing **120** (a phase separation partition is disposed between adjacent circuit breaker poles **300** in the case that the circuit breaker pole **300** is not provided with a unit housing **120**, and the slide rail **25** is disposed on a sidewall of the phase separation partition), and the slide rail **25** penetrates through the bracket **50** or the unit housing **120** (or the phase separation partition) in a thickness or depth direction. Further, as shown in FIGS. 16 and 22, two ends of the slider **26** pass through the two slide rails **25** respectively; one end of the first connecting rod **27**, which is connected to the slider **26**, is located between the two slide rails **25**; and one ends of the two second connecting rods **29**, which are connected to the slider **26**, are located on two sides of the two slide rails **25** respectively, and rotatably connected to two ends of the slider **26** respectively.

(66) Further, each slide rail **25** is straight, arc-shaped, triangular, or a combined shape of straight line and arc. Further, the shape of the slide rail **25** may also be adaptively set according to a current level of the circuit breaker, a design space, control requirements, and the like. As shown in FIGS.

11, 12 and 13, the slider **26** moves upward along the slide rails **25** in response to a disconnecting or tripping action of the circuit breaker; and the slider **26** moves downward along the slide rails **25** when the circuit breaker is closed. A trajectory of upward movement and a trajectory of downward movement are coincident, which, in this case, may be set to be straight, arc-shaped, or a combined shape of straight line and arc. For example, the slide rail **25** is set in a triangular shape or other shapes. In the case that the slide rail is set as a triangle, a movement trajectory of the slider **26** along the slide rails **25** is upward along one side of the triangle when the circuit breaker is disconnected or tripped; the movement trajectory of the slider **26** along the slide rails **25** is downward along the other side of the triangle, but the downward and upward movement trajectories are not coincident (not shown); the slider **26** forms a closed-shaped movement trajectory along the slide rails **25**.

(67) As shown in FIGS. **11-13, 23** and **46**, an embodiment of the slide rails **25** and the slider **26** is shown. The bracket **50** includes two bracket arms **501** disposed to face with other at intervals, and each bracket arm **501** is provided with a slide rail **25**, and two ends of the slider **26** are respectively slidably disposed on the two slide rails **25**. Further, as shown in FIGS. **7-13, 23** and **46**, each slide rail **25** is a slide hole, and the slider **26** is a sliding shaft, two ends of which are respectively disposed in the two slide holes. Further, as shown in FIGS. **7-13, 23, 41** and **46**, each slide rail **25** is a straight hole, the slider **26** is a sliding shaft disposed in the straight hole, and two ends of the sliding shaft are provided with sliding shaft grooves which are in limiting fit with the sidewall of the straight hole.

(68) Further, as shown in FIG. **11**, the other end of the second connecting rod **29** is rotatably connected to the contact support **110** directly through a first connecting shaft **21**. Further, as shown in FIG. **47**, the bracket **50** further includes a second avoidance hole **509** for the first connecting shaft **21** to pass through for avoiding the first connecting shaft **21**. Further, as shown in FIG. **47**, the second avoidance hole **509** is an arc-shaped hole, which is matched with a movement trajectory of the first connecting shaft **21**.

(69) Further, as shown in FIG. **41**, the operating mechanism **100** includes a plurality of contact supports **110** arranged side by side at intervals, and the respective contact supports **110** are linked by means of a linkage shaft **5**; and the housing of the circuit breaker is provided with a first avoidance hole **120-7** for the linkage shaft **5** to pass through for avoiding the linkage shaft **5**. Further, as shown in FIG. **41**, the first avoidance hole **120-7** is an arc-shaped hole, which is matched with a movement trajectory of the linkage shaft **5**. Further, as shown in FIG. **41**, the housing of the circuit breaker includes a unit housing **120**, in which the first avoidance hole **120-7** is formed; or in the case that the housing of the circuit breaker is provided with a phase separation partition, the first avoidance hole is formed in the phase separation partition.

(70) Further, as shown in FIGS. **1-13**, the operating mechanism **100** further includes a latch **13** and a re-buckle **15** pivotally disposed on the bracket **50** respectively, wherein a jump buckle **60** is in latching fit with the latch **13**, and the latch **13** is in limiting fit with the re-buckle **15**. Specifically, as shown in FIGS. **1-6** and **25**, the jump buckle **60** is provided with a jump buckle hasping surface **604** at one end, and the latch **13** is provided with a latch hasping surface, wherein the jump buckle hasping surface **604** is located below the latch hasping surface and is in limiting fit with the latch hasping surface to realize latching fit of the jump buckle **60** and the latch **13**. As shown in FIGS. **1-6**, one end of the re-buckle **15** is located on one side of the latch **13** and limiting fit with the latch **13**; the re-buckle **15** releases the limiting fit from the latch **13** while the re-buckle **15** is driven to rotate; and the latch **13** rotates and releases the limiting fit of the jump buckle hasping surface **604** and the latch hasping surface so that the latch **13** releases the latching fit from the jump buckle **60**. Further, the circuit breaker of the present invention further includes an overload and short-circuit protection mechanism. The overload and short-circuit protection mechanism will drive the re-buckle **15** to rotate while a short-circuit or overload fault occurs in the circuit breaker, so that the re-buckle **15** releases the limiting fit from the latch **13**. Further, the short-circuit and overload

protection mechanism includes a short-circuit protection mechanism and an overload protection mechanism, wherein the short-circuit protection mechanism is preferably an electromagnetic trip, and the overload protection mechanism is preferably a thermal tripping mechanism (e.g., the overload protection mechanism includes a bimetal sheet). It should be pointed out that the cooperation between the latch **13** and the jump buckle **60**, the cooperation between the latch **13** and the re-buckle **15**, and the cooperation between the re-buckle **15** and the short-circuit and overload protection mechanism may all be realized by the prior art, which will not be expanded in detail.

(71) Further, as shown in FIGS. **1-13**, the rocker arm assembly includes a synchronously acting handle **41**, a rocker arm **45** fixedly connected to the handle **41**, and a reset structure **42** for driving the jump buckle **60** to rotate to re-buckle with the latch **13**, wherein the rocker arm **45** is pivotally disposed on the bracket **50**, and two ends of a swing stroke of the rocker arm **45** on the rocker arm assembly are in limiting fit with the bracket **50**, respectively. Further, as shown in FIGS. **4-6**, and **11-16**, the reset structure **42** is a reset shaft; the jump buckle **60** is of a strip structure, wherein one end of the jump buckle **60** is in latching fit with the latch **13**, and the other end of the jump buckle **60** is pivotally disposed on the bracket **50**; and the jump buckle **60** includes a driving side edge **603** disposed on one side to be in driving fit with the reset structure **42**. Specifically, in the direction shown in FIGS. **4-6** and **11-16**, the driving side edge **603** is disposed at the upper side edge of the jump buckle **60**.

(72) Further, as shown in FIGS. **1-13**, the first crank **30** includes a crank limiting portion **31**. The crank limiting portion **31** is in limiting fit with the jump buckle **60** in the case that the circuit breaker is in a closed state or tripped state. Further, as shown in FIGS. **4-6**, and **11-16**, the crank limiting portion **31** is a limiting shaft, and is in limiting fit with the jump buckle **60** to prevent the first crank **30** from rotating. Further, as shown in FIGS. **4-6** and **11-16**, the jump buckle **60** includes a limiting side edge **608** disposed at one side edge to cooperate with the crank limiting portion **31**, and the limiting side edge **608** is an arc-shaped side edge. Specifically, as shown in FIGS. **4-6** and **11-16**, the limiting side edge **608** is disposed at the lower side edge of the jump buckle **60**.

(73) Further, as shown in FIGS. **1-13**, the first connecting rod **27** and the first crank **30** are rotatably connected around a second axis **16m**; one end of the first spring **22** is rotatably connected to the second axis **16m**, and the other end of the first spring **22** is rotatably disposed on the rocker arm assembly around a fourth axis **46m**. Further, as shown in FIGS. **1-13**, the first crank **30** and the first connecting rod **27** are rotatably connected by a second spring shaft **16**, one end of the first spring **22** is connected to the second spring shaft **16**, and the other end of the first spring **22** is connected to the rocker arm **45** through the first spring shaft **46**.

(74) Further, as shown in FIGS. **1-3**, the jump buckle **60** is pivotally disposed on the bracket **50** around a fifth axis **11s**; the re-buckle **15** is pivotally disposed on the bracket **50** around a sixth axis **14s**; the latch **13** is pivotally disposed on the bracket **50** around a seventh axis **12s**; the rocker arm **45** is pivotally disposed on the bracket **50** around an eighth axis **28s**; and one end of the first spring **22** is pivotally disposed on the rocker arm **45** around a fourth axis **46**. Further, as shown in FIGS. **4-6**, the jump buckle **60** is pivotally disposed on the bracket **50** through a jump buckle shaft **11**; the re-buckle **15** is pivotally disposed on the bracket **50** through a re-buckle shaft **14**; the latch **13** is pivotally disposed on the bracket **50** through a latch shaft **12**; the rocker arm **45** is pivotally disposed on the bracket **50** through a rocker arm shaft **28**; two ends of the first spring **22** are rotatably connected to the rocker arm **45** and the first crank **30** through the first spring shaft **46** and the second spring shaft **16** respectively; the first spring shaft **46** and the second spring shaft **16** are respectively located on two sides of the jump buckle **60**, and the first crank **30** is pivotally disposed on the jump buckle **60** through a first crank shaft **67**; and one end of the first connecting rod **27** is rotatably connected to the first crank **30** through the second spring shaft **16**, and the other end of the first connecting rod **27** is rotatably connected to the slider **26**. Further, as shown in FIG. **22**, each bracket arm **501** of the bracket **50** is provided with a bracket jump buckle shaft hole **511**, a bracket-re-buckle shaft hole **514**, a bracket-latch shaft hole **512** and a bracket-rocker arm shaft

groove **528** which cooperate with the jump buckle shaft **11**, the re-buckle shaft **14**, the latch shaft **12** and the rocker arm shaft **28**, respectively. As shown in FIG. **25**, the jump buckle **60** includes a jump buckle shaft hole **601** disposed in one end thereof, the first crank **30** is provided with a crank shaft hole **307** that cooperates with the first crank shaft **67**, and a jump buckle-crank shaft hole that cooperates with the first crank shaft **67** is formed in the middle of the jump buckle **60**.

(75) Further, as shown in FIGS. **4-6**, **11-16**, and **25**, an embodiment of the first crank **30** is shown: the first crank **30** is of a triangular structure, in which one vertex is pivotally disposed on the jump buckle **60** around the first axis **67m**, another vertex is rotatably connected to the first spring **22** and the first connecting rod **27** around the second axis **16m** respectively, and the third vertex is provided with a crank limiting portion **31**. Further, as shown in FIG. **26**, two first cranks **30** are respectively disposed on two sides of the jump buckle **60**, and the three vertices of the two first cranks **30** are connected to the crank limiting portion **31** respectively through the first crank shaft **67**, the second spring shaft **16** and the crank limiting portion **31**. Further, as shown in FIG. **25**, the first crank **30** includes a crank shaft hole **307**, a limiting shaft hole **301**, and a crank-spring shaft groove **302** which cooperate with the first crank shaft **67**, the crank limiting portion **31** and the second spring shaft **16** respectively, and the crank shaft hole **307**, the limiting shaft hole **301** and the crank-spring shaft groove **302** are located in three vertices of the first crank **30** respectively.

(76) Further, as shown in FIGS. **4-6** and **11-16**, the bracket **50** includes a V-shaped groove **505**, wherein the rocker arm **45** is in limiting fit with two sidewalls of the V-shaped groove **505** at the first stroke end and the second stroke end respectively. Further, as shown in FIGS. **22** and **23**, the bracket **50** comprises two bracket arms **501** disposed to face each other at intervals and a bracket connecting plate **502**, wherein two ends of the bracket connecting plate **502** are respectively connected to the two bracket arms **501** by bending, so that the bracket **50** is integrally of a U-shaped structure, and each support arm **501** is provided with a V-shaped groove **505**. As shown in FIGS. **27** and **28**, the rocker arm **45** includes a pair of rocker arm legs **408** disposed to face each other at intervals, which are respectively disposed in two V-shaped grooves **505** and are rotatably connected to the two bracket arms **501** respectively.

(77) Specifically, as shown in FIGS. **4-6** and **11-16**, the lower end of each rocker arm leg **408** is pivotally disposed at the bottom of the V-shaped groove **505** through the rocker arm shaft **28**; and at the first stroke end and the second stroke end of the rocker arm, and the rocker arm legs **408** are in limiting fit with a right sidewall and a left sidewall of the V-shaped groove, respectively. Further, as shown in FIG. **22**, a bracket-rocker arm shaft groove **528** which cooperates with the rocker arm shaft **28** is formed in the bottom of the V-shaped groove **505**. As shown in FIG. **27**, a rocker arm shaft groove **428** which cooperates with the rocker arm shaft **28** and oppositely cooperates with the bracket-rocker arm shaft groove **528** is formed in one end of the rocker arm leg **408**.

(78) The following is a process of switching the operating mechanism **100** in the first embodiment among a disconnected state, a closed state and a tripped state, and the details are as follows:

(79) as shown in FIGS. **7-13**, two ends of the swing stroke of the rocker arm **45** are the first stroke end and the second stroke end respectively; and two ends of the first spring **22** are the first spring end **220** and the second spring end **221** respectively, which are connected with the rocker arm assembly and the first crank **30** respectively. Specifically, as shown in FIGS. **7-13**, the first stroke end and the second stroke end of the rocker arm **45** are a right end and a left end of the swing stroke of the rocker arm **45** respectively, the upper end of the first spring **22** is the first spring end **220**, and the lower end of the first spring **22** is the second spring end **221**.

(80) The operation process of the operating mechanism **100** being switched from the closed state to the disconnected state will be described below in conjunction with FIGS. **7**, **8**, **11** and **12**: as shown in FIG. **7** and FIG. **11**, in the case that the operating mechanism **100** is in a closed state, the rocker arm **45** swings toward the second stroke end and drives the first spring end **220** to rotate around the second spring end **221**, until the first spring **22** passes a first dead center position; the first spring **22** drives the first crank **30** to rotate in a second direction and drives the rocker arm **45** to swing to the

second stroke end; the first crank **30** drives the slider **26** through the first connecting rod **27** to slide along the slide rails **25**; and the slider **26** drives the contact support **110** through the second connecting rod **29** to rotate in the first direction to a breaking position (the moving contact **9** and the static contact **18** are in a disconnected state while the contact support **110** is located in the breaking position), so that the operating mechanism is switched to a disconnected state shown in FIG. **8** and FIG. **12**. Specifically, as shown in FIGS. **7**, **8**, **11** and **12**, in the case that the operating mechanism **100** is switched from a closing position to a disconnecting position, the slider **26** moves upward along the slide rails **25**, the first direction is counterclockwise, and the second direction is clockwise. When the first spring **22** is located at the first dead center position, the energy storage of the first spring **22** reaches a maximum value, the first axis **67m** is located on a first axis line, and the first axis **22** passes the first axis **67m** while the first spring **22** passes the first dead center position around the second spring end **221**. Therefore, the first axis **67m** may also be regarded as the first dead center position, that is to say, the first axis line passes the first axis **67m**, i.e., the first spring **22** passes the first dead point position. It should be pointed out that, as shown in FIG. **12**, the contact support **110** and/or the moving contact **9** are limited by the unit housing **120** while the operating mechanism **100** is in the disconnected state, so that the contact support **110** cannot continue to rotate in the first direction, and the contact support **110** forms limiting for the slider **26** through the second connecting rod **29** at the same time, thereby preventing the slider **26** from sliding upward along the slide rails **25**.

(81) The operation process of the operating mechanism **100** being switched from the disconnected state to the closed state will be described below in conjunction with FIGS. **7**, **8**, **11** and **12**: as shown in FIG. **8** and FIG. **12**, in the case that the operating mechanism **100** is in the disconnected state, the rocker arm **45** swings toward the first stroke end and drives the first spring end **220** to rotate around the second spring end **221**, until the first spring **22** passes the first dead center position; the first spring **22** drives the first crank **30** to rotate in the first direction, such that the crank limiting portion **31** is in limiting fit with the jump buckle **60**, thereby preventing the first crank **30** from rotating in the first direction; meanwhile, the first spring **22** drives the rocker arm **45** to swing to the first stroke end, and the first crank **30** drives the slider **26** through the first connecting rod **27** to slide along the slide rails **25**; and the slider **26** drives the contact support **110** through the second connecting rod **29** to rotate in the second direction to the closing position (the moving contact **9** and the static contact **18** are in a closed state while the contact support **110** is located in the closing position), so that the operating mechanism is switched to the closed state; and the first direction and the second direction are opposite to each other. Specifically, as shown in FIGS. **7**, **8**, **11** and **12**, the slider **26** moves downward along the slide rails **25** while the operating mechanism **100** is switched from the disconnected state to the closed state. It should be pointed out that, as shown in FIG. **7**, the moving contact **9** and the static contact **18** are closed while the operating mechanism **100** is in the closed state, so that the contact support **110** cannot continues to rotate in the second direction, and the contact support **110** forms limiting for the slider **26** through the second connecting rod **29** at the same time, thereby preventing the slider **26** from sliding downward along the slide rails **25**.

(82) The operation process of the operating mechanism **100** being switched from the closed state to the tripped state will be described below in conjunction with FIGS. **7**, **9**, **11** and **13**: as shown in FIG. **7** and FIG. **11**, in the case that the operating mechanism **100** is in the closed state, the re-buckle **15** rotates to release the limiting fit from the latch **13**, and the latch **13** rotates to release the latching fit from the jump buckle **60**; the jump buckle **60** rotates and drives the first crank **30** to rotate synchronously, the first crank **30** drives the slider **26** through the first connecting rod **27** to slide along the slide rails **25**, and meanwhile, the slider **26** drives the contact support **110** through the second connecting rod **29** to rotate in the second direction to a breaking position; and the first spring **22** drives the rocker arm **45** to swing toward the second stroke end to the reset structure **42** to be in limiting fit with the jump buckle **60**, and the operating mechanism is switched to the

tripped state as shown in FIG. 9 and FIG. 13. Specifically, as shown in FIGS. 7, 9, 11 and 13, the slider **26** moves upward along the slide rails **25** while the operating mechanism **100** is switched from the closed state to the tripped state. It should be pointed out that, as shown in FIG. 13, the contact support **110** and/or the moving contact **9** are limited by the unit housing **120** while the operating mechanism **100** is in the tripped state, so that the contact support **110** cannot continue to rotate in the first direction, and the contact support **110** forms limiting for the slider **26** through the second connecting rod **29** at the same time, thereby preventing the slider **26** from sliding upward along the slide rails **25**.

(83) The operation process of the operating mechanism **100** being switched from the tripped state to the disconnected state will be described below in conjunction with FIGS. 8, 9, 11 and 13: as shown in FIG. 9 and FIG. 13, in the case that the operating mechanism **100** is in the tripped state, the rocker arm **45** swings to the second stroke end, and the rocker arm **45** drives the jump buckle **60** through the reset structure **42** to rotate to be in latching fit with the latch **13**; and meanwhile, the latch **13** rotates to be in limiting fit with the re-buckle **15**, and the operating mechanism is switched to the disconnected state shown in FIG. 8 and FIG. 11.

(84) The following is a second embodiment of the operating mechanism **100**, to be specific: the operating mechanism **100** of the second embodiment differs from the operating mechanism **100** of the first embodiment in that: the operating mechanism **100** in the second embodiment further includes an auxiliary limiting structure, wherein one end of the auxiliary limiting structure is rotatably connected to the second connecting rod **29**, and the other end of the auxiliary limiting structure is rotatably connected to the bracket **50** or a housing of the circuit breaker. The auxiliary limiting structure cooperates with the movement of the slider **26** to affect a movement state of the second connecting rod **29**, thereby jointly defining a movement trajectory of the rotating connection position of the second connecting rod **29** and the contact support **100**.

(85) It should be pointed out that the auxiliary limiting structure is rotatably connected to one end of the bracket **50** or the housing of the circuit breaker, or may also be configured to reciprocate along a predetermined trajectory, for example, this end is slidably disposed in a track. Further, in the case that one end of the auxiliary limiting structure is connected to the housing of the circuit breaker, this end may be connected to the circuit breaker shell **3** or the unit housing **120** (or the phase separation partition).

(86) Further, referring to FIGS. 48 and 49, the auxiliary limiting structure is of a connecting rod structure or a crank-slider structure. Further, the auxiliary limiting structure is a third crank **204**, wherein one end of the third crank **204** is rotatably connected to the second connecting rod **29**, and the other end of the third crank **204** is rotatably connected to the bracket **50** or the housing of the circuit breaker. The second connecting rod **29** moves accordingly while the slider **26** slides along the slide rails **25**, and at the same time, the third crank **204** rotates around a part of the third crank **204** that is rotatably disposed on the bracket **50** or the housing of the circuit breaker to assist in limiting the movement of the second connecting rod **29**; and the second connecting rod **29** and the third crank **204** cooperate to make the movement of the second connecting rod **29** more precise.

(87) Further, the third crank has a straight or arc-shaped plate structure.

(88) As shown in FIG. 7-13 and FIG. 47, a third embodiment of the operating mechanism **100** is specifically as follows: the operating mechanism **100** of the third embodiment differs from the operating mechanism **100** of the first embodiment in that: the operating mechanism **100** further includes a second crank **19**, wherein the second crank **19** includes a second crank supporting portion **206**, a second crank connecting portion **208**, and a second crank driving portion **210**; the second crank **19** is pivotally disposed through the second crank supporting portion **206**, and the second crank **19** is connected to the contact support **110** through the second crank driving portion **210**; one end of the second connecting rod **29** is rotatably connected to the slider **26**, and the other end of the second connecting rod **29** is rotatably connected to the second crank connecting portion **208**; and the contact support **110** is driven to rotate around the third axis **111s**. Further, the second

crank **19** is pivotally disposed on the bracket **50** or the housing of the circuit breaker through the second crank supporting portion **208**. Further, the rocker arm assembly can drive the first crank **30** to swing through the first spring **22**, the first crank **30** drives the slider **26** through the first connecting rod **27** to slide on the slide rails **25**, the slider **26** drives the second crank **19** to swing through the second connecting rod **29**, the second crank **19** drives the contact support **110** to rotate, and the contact support **110** drives the moving contact **9** of the circuit breaker to rotate. The slider **26** slides on the slide rails **25**, and the slider **26** drives the second crank **19** to swing through the second connecting rod **29**. A distance between the third axis **111s** and a rotation axis line (referring to a reference numeral **21** in FIGS. **11-14**) at the rotating connection position of the second connecting rod **29** and the second crank **19** is much smaller than the length of the moving contact **9**, so the second connecting rod **29** drives the second crank **19** to generate a small rotation, which will be proportionally enlarged into an opening distance between the moving contact and the static contact. The opening distance of the moving contact **9** can be adjusted by adjusting a connection position between the second connecting rod **29** and the second crank **19**.

(89) Specifically, in the case that the second crank supporting portion **206** of the second crank **19** is disposed on the housing of the circuit breaker, the second crank supporting portion **206** may be pivotally disposed on the circuit breaker housing **3** or the unit housing **120** (or the phase separation partition). Further, as shown in FIG. **41**, the second crank supporting portion **206** is pivotally disposed on the unit housing **120** through a second crank shaft **79**. Further, the unit housing **120** includes a second crank shaft hole which is a blind hole, and the second crank shaft **79** cooperates with the second crank shaft hole.

(90) Specifically, as shown in FIG. **47**, in the case that the second crank supporting portion **206** of the second crank **19** is disposed on the bracket **50**, the second crank supporting portion **206** is pivotally disposed on the sidewall of the bracket **50** through the second crank shaft **79**. The second crank shaft **79** may be a common connector such as a rivet, a screw rod or a bolt. The second crank supporting portion **206** is pivotally disposed on the bracket **50**, which is beneficial to further reduce the assembly error of the operating mechanism **100** and improve the operational reliability of the operating mechanism **100**.

(91) Further, as shown in FIGS. **11-16**, **41** and **47**, the second crank connecting portion **208** of the second crank **19** is disposed between the second crank supporting portion **206** and the second crank driving portion **210**. Specifically, as shown in FIGS. **11-13**, the second crank supporting portion **206** and the second crank driving portion **210** are respectively disposed on two ends of the second crank **19**; and the second crank supporting portion **208** is disposed in the middle of the second crank **19** and located between the second crank supporting portion **206** and the second crank driving portion **210**.

(92) Further, the center of rotation of the second crank **19** is a ninth axis, and the ninth axis is parallel to or coincided with the third axis **111s**. Specifically, as shown in FIG. **41**, the ninth axis coincides with the third axis **111s** in the case that the second crank supporting portion **206** is pivotally arranged on the unit housing **120**. As shown in FIG. **47**, the second crank supporting portion **206** is pivotally disposed on the bracket **50**, and the ninth axis is parallel to, but not coincident with the third axis **111s**.

(93) Further, as shown in FIGS. **11-16** and **41**, the second connecting rod **29** is rotatably connected to the second crank connecting portion **208** of the second crank **19** through the first connecting shaft **21**.

(94) Further, as shown in FIGS. **11-16** and **41**, the second crank driving portion **210** of the second crank **19** is driving connection to the contact support **110** through the linkage shaft **5**. The linkage shaft **5** is a connecting shaft for realizing the synchronous rotation of the respective contact supports **110**, and the second crank driving portion **210** of the second crank **19** is connected with the linkage shaft **5** to drive the contact support **110** to rotate, which is beneficial to improve the action synchronicity of respective circuit breaker poles **300**.

(95) Further, the second connecting rod **29** has an arc or straight plate structure, wherein one end of the second connecting rod **29** is rotatably connected to the slider **26** and the other end of the second connecting rod **29** is rotatably connected to the second crank **19** (or, in the operating mechanism **100** of the first embodiment, the second crank **19** is directly connected to the contact support **110**). Further, as shown in FIGS. **11-16**, **41** and **47**, the second connecting rod **29** has an arc-shaped plate structure, wherein one end of the second connecting rod **29** is rotatably connected to the slider **26**, and the other end of the second connecting rod **29** is rotatably connected to the second crank connecting portion **208** of the second crank **19**. It should be pointed out that the shape of the second connecting rod **29** can be adaptively set according to specific space conditions, and can be designed into a shape that meets the requirements when it is necessary to avoid a specific structure, wherein the arc shape or the straight plate shape is a relatively conventional design, but the shape of the second connecting rod **29** is not limited to the arc or straight plate structure.

(96) As shown in FIG. **1-13**, a fourth embodiment of the operating mechanism **100** is specifically as follows: the operating mechanism **100** of the fourth embodiment differs from the operating mechanisms **100** of the first to third embodiments in that: as shown in FIGS. **2-3**, **5-6**, **8-9**, and **12-13**, the slide rails **25** are defined on the bracket **50**, and the slider **26** is in limiting fit the slide rails **25** to prevent the slider **26** from sliding while the operating mechanism **100** is in a disconnected state or a tripped state. The slide rails **25** provide a guiding function for the slider **26**, and also serve as supporting points to provide a supporting force for the first connecting rod **27** and the slider **26**, so that the operating mechanism **100** may have a stable closing position, disconnecting position and tripping position without cooperating with the contact support **110**, making the operating mechanism **100** become an independently operable mechanism, which is conducive to the modularized assembly and production of the operating mechanism **100**, and provides more design space for the distribution of the operating mechanism **100** in the circuit breaker shell **3**. In addition, in actual production, the operating mechanism **100** does not need to cooperate with the contact system, which avoids the loss of the contact system during the test process, improves the assembly efficiency, and helps reduce R&D and production cost. Further, the operating mechanism **100** of the fourth embodiment can be independently switched among the closed state, the disconnected state and the tripped state in the case that the second connecting rod **29**, the contact support **110** and the moving contact **9** are detached.

(97) As shown in FIGS. **4-6** and **11-16**, a layout form of the operating mechanism **100** of the fourth embodiment is as follows: the re-buckle **15**, the latch **13**, the jumping pin **16** and the first crank **30** are all arranged between two bracket arms **501**; one end of the jump buckle **60** is pivotally arranged on the bracket connecting plate **502**, and the other end of the jump buckle **60** is in latching fit with the latch **13**; the re-buckle **15** and the latch **13** are disposed on one side of the V-shaped groove **505**, and the bracket connecting plate **502** is located on the other side of the V-shaped groove **505**; one end of the rocker arm leg **408** is pivotally disposed at the bottom of the V-shaped groove **505**; one end of the first crank **30** is rotatably connected to the middle part of the jump buckle **60**, the other end of the first crank **30** is rotatably connected to one end of the first connecting rod **27**, and the other end of the first connecting rod **27** is in driving connection with the slider **26**; and the slide rails **25** are disposed on the bracket arms **501**, and the slide rails **25** and the V-shaped groove **5** are respectively disposed on two ends of the bracket arms **501** and have opposite opening directions. Further, as shown in FIGS. **7-16**, the slider **26** is further connected to one end of the second connecting rod **29**, and the other end of the second connecting rod **29** is connected to the second crank **19**; one end of the second crank **19** is pivotally disposed around the ninth axis, the ninth axis coincides with the third axis **111s**, and the other end of the second crank **19** is connected to the contact support **110**.

(98) Specifically, in the direction shown in FIGS. **4-6** and **11-16**, the right end of the jump buckle **60** is pivotally arranged on the bracket connecting plate **502**, and the left end of the jump buckle **60** is in latching fit with the latch **13**; the re-buckle **15** and the latch **13** are disposed on the left side of

the V-shaped groove **505**, and the bracket connecting plate **502** is located on the right side of the V-shaped groove **505**; the lower end of the rocker arm leg **408** is pivotally disposed at the bottom of the V-shaped groove **505**; the upper end of the first crank **30** is rotatably connected to the middle part of the jump buckle **60**, the lower end of the first crank **30** is rotatably connected to the upper end of the first connecting rod **27**, and the lower end of the first connecting rod **27** is in driving connection with the slider **26**; and the slide rails **25** and the V-shaped groove **505** are disposed at the lower end and the upper end of the bracket arm **501** respectively, and have openings which face downward and upward respectively. Further, in a direction shown in FIGS. 7-16, the upper end of the second connecting rod **29** is connected to the slider **26**, the lower end of the second connecting rod **29** is connected to the second crank **19**, the upper end of the second crank **19** and the contact support **110** are pivotally disposed around the third axis **111s** respectively, and the lower end of the second crank **19** is in driving connection to the contact support **110**.

(99) In order to better illustrate the structure and principle of the operating mechanism **100**, a cooperative relationship of respective components of the operation mechanism **100** of the fourth embodiment are described in detail below in three states (closed, disconnected and tripped states), to be specific:

(100) As shown in FIGS. 1-6, two ends of the swing stroke of the rocker arm **45** are a first stroke end and a second stroke end, respectively; two ends of the first spring **22** are a first spring end **220** and a second spring end **221** respectively, which are connected to the rocker arm assembly and the first crank **30** respectively; and an axis line of the first spring **22** is a first axis line, a first axis line side and a second axis line side are respectively located on two sides of the first axis line. As shown in FIG. 1 and FIG. 4, in the case that the operating mechanism **100** is in the closed state, the rocker arm **45** is located at the first stroke end, the re-buckle **15** is in limiting fit with the latch **13**, the latch **13** is in latching fit with the jump buckle **60**, the crank limiting portion **31** is in limiting fit with the jump buckle **60** to prevent the first crank **30** from rotating in the first direction, and the first axis **67m** is located at the first axis line side. As shown in FIG. 2 and FIG. 5, in the case that the operating mechanism **100** is in the disconnected state, the rocker arm **45** is located at the second stroke end, the re-buckle **15** is in limiting fit with the latch **13**, the latch **13** is in latching fit with the jump buckle **60**, the crank limiting portion **31** releases the limiting fit from the jump buckle **60**, the reset structure **42** is in limiting fit with the jump buckle **60**, and the slider **26** is in limiting fit with the slide rails **25** to prevent the first crank **30** from rotating in the second direction through the first connecting rod **27**. The first and second directions are opposite to each other, and the first axis **67m** is located at the second axis line side. Further, as shown in FIGS. 3 and 6, in the case that the operating mechanism **100** is in the tripped state, the rocker arm **45** is located in the middle of its swing stroke, the re-buckle **15** releases the limiting fit from the latch **13**, the latch **13** releases the latching fit from the jump buckle **60**, the crank limiting portion **13** is in limiting fit with the jump buckle **60**, the reset structure **42** is in limiting fit with the jump buckle **60**, the slider **26** is in limiting fit with the slide rails **25**, and the first axis **67m** is located at the first axis line side. The operating mechanism **100** enters the disconnected state after re-buckle from the tripped state.

(101) Specifically, in a direction shown in FIGS. 1-6, the first stroke end is the right end of the swing stroke of the rocker arm assembly or rocker arm **45**, and the second stroke end is the left end of the swing stroke of the rocker arm assembly or rocker arm **45**; the first axis line side is the left side of the first axis line, and the second axis line side is the right side of the first axis line; and the first direction is counterclockwise and the second direction is clockwise.

(102) It should be pointed out that the “re-buckle” of the operating mechanism **100** refers that the latch **13** and the jump buckle **60** restore the latching fit, and the re-buckle **15** and the latch **13** restore the limiting fit.

(103) The following is an operation process of switching the operating mechanism **100** of the fourth embodiment among a closed state, a disconnected state and a tripped state: the operation process of the operating mechanism **100** being switched from the closed state to the disconnected

state will be described below in conjunction with FIGS. 1, 4, 2 and 5: as shown in FIG. 1 and FIG. 4, in the case that the operating mechanism **100** is in the closed state, the rocker arm **45** swings toward the second stroke end and drives the first spring end **220** to rotate around the second spring end **221**, until the first spring **22** passes a first dead center position; the first spring **22** drives the first crank **30** to rotate in the second direction and drives the rocker arm **45** to swing to the second stroke end; the first crank **30** drives the slider **26** through the first connecting rod **27** to move till being in limiting fit with the slide rails **25** so as to prevent the first crank **30** from rotating in the second direction, so that the operating mechanism **100** is switched to the disconnected state shown in FIG. 2 and FIG. 5. Specifically, in a direction shown in FIG. 1 and FIG. 4, in the case that the operating mechanism **100** is in the closed state, the rocker arm **45** swings from right to left (from the first stroke end to the second stroke end) and drives the first spring end **220** to rotate around the second spring end **221** in a anticlockwise direction, until the first spring **22** passes the first dead center position; the first spring **22** drives the first crank **30** to rotate in a clockwise direction (the second direction) and drives the rocker arm **45** to swing to the second stroke end rapidly; the first crank **30** drives the slider **26** through the first connecting rod **27** to move to the upper end of the slide rails **25** and to be in limiting fit therewith, thereby preventing the first crank **30** from rotating in the clockwise direction (the second direction), so that the operating mechanism **100** is switched to the disconnected state shown in FIG. 2 and FIG. 5.

(104) The operation process of the operating mechanism **100** being switched from the disconnected state to the closed state will be described below in conjunction with FIGS. 1, 4, 2 and 5: as shown in FIG. 2 and FIG. 5, in the case that the operating mechanism **100** is in the disconnected state, the rocker arm **45** swings toward the first stroke end and drives the first spring end **220** to rotate around the second spring end **221**, until the first spring **22** passes the first dead center position; the first spring **22** drives the first crank **30** to rotate in the first direction, such that the crank limiting portion **31** is in limiting fit with the jump buckle **60**, thereby preventing the first crank **30** from rotating in the first direction; meanwhile, the first spring **22** drives the rocker arm **45** to swing to the first stroke end, so that the operating mechanism **100** is switched to the closed state as shown in FIG. 1 and FIG. 4; and the first direction and the second direction are opposite to each other. Specifically, in a direction shown in FIG. 2 and FIG. 5, in the case that the operating mechanism **100** is in the disconnected state, the rocker arm **45** swings from left to right (from the second stroke end to the first stroke end) and drives the first spring end **220** to rotate around the second spring end **221** in a clockwise direction, until the first spring **22** passes the first dead center position; the first spring **22** drives the first crank **30** to rotate in an anticlockwise direction (the first direction), such that the crank limiting portion **31** is in limiting fit with the jump buckle **60**, thereby preventing the first crank **30** from continuing to rotate in the anticlockwise direction (the first direction); meanwhile, the first spring **22** drives the slider **26** through the first connecting rod **27** to move to the middle part of the slide rails **25** from the upper end of the slide rails **25**, and meanwhile, drives the rocker **45** to swing to the first stroke end rapidly, so that the operating mechanism **100** is switched to the closed state shown in FIG. 1 and FIG. 4.

(105) The operation process of the operating mechanism **100** being switched from the closed state to the tripped state will be described below in conjunction with FIGS. 1, 4, 3 and 6: as shown in FIG. 1 and FIG. 4, in the case that the operating mechanism **100** is in the closed state, the re-buckle **15** rotates to release the limiting fit from the latch **13**, and the latch **13** rotates to release the latching fit from the jump buckle **60**; the jump buckle **60** rotates and drives the first crank **30** to rotate synchronously, the first crank **30** drives the slider **26** through the first connecting rod **27** to move till being in limiting fit with the slide rails **25**, thereby preventing the jump buckle **60** from continuing to rotate; and the first spring **22** drives the rocker arm **45** to swing toward the second stroke end to the reset structure **42** to be in limiting fit with the jump buckle **60**, such that the operating mechanism is switched to the tripped state as shown in FIG. 3 and FIG. 6. Specifically, in the direction shown in FIG. 1 and FIG. 4, in the case that the operating mechanism **100** is in the

closed state, the re-buckle **15** rotates anticlockwise to release the limiting fit from the latch **13**, and the latch **13** rotates anticlockwise to release the latching fit from the jump buckle **60**; the jump buckle **60** rotates clockwise and drives the first crank **30** to rotate synchronously, the first crank **30** drives the slider **26** through the first connecting rod **27** to slide to the upper end of the slide rails **25** and to be in limiting fit therewith, thereby preventing the jump buckle **60** from continuing to rotate clockwise; and the first spring **22** drives the rocker arm **45** to swing anticlockwise (in a direction where the second stroke end is located) to the reset structure **42** and be in limiting fit with the jump buckle **60**, such that the operating mechanism **100** is switched to the tripped state shown in FIG. 3 and FIG. 6.

(106) The operation process of the operating mechanism **100** being switched from the tripped state to the disconnected state will be described below in conjunction with FIGS. 3, 6, 2 and 5: as shown in FIG. 3 and FIG. 6, in the case that the operating mechanism **100** is in the tripped state, the rocker arm **45** swings to the first stroke end, and the jump buckle **60** is driven by the reset structure **42** to rotate to be in latching fit with the latch **13**, such that the latch **13** is in limiting fit with the re-buckle **15**, and the operating mechanism **100** is switched to the disconnected state shown in FIG. 2 and FIG. 5. Specifically, in the direction shown in FIG. 3 and FIG. 6, in the case that the operating mechanism **100** is in the tripped state, the rocker arm **45** swings to the first stroke end anticlockwise, and the jump buckle **60** is driven by the reset structure **42** to rotate to be in latching fit with the the latch **13**, and meanwhile, the jump buckle **60** drives the latch **13** to rotate clockwise, such that the latch **13** is in limiting fit with the re-buckle **15**, and the operating mechanism **100** is switched to the disconnected state shown in FIG. 2 and FIG. 5.

(107) Further, as shown in FIGS. 1-6, the first spring **22** is located in the first dead center position, and the first axis **67m** is located on the first axis line. Further, as shown in FIGS. 1-6, the first spring **22** is a tension spring, the first direction refers to a direction facing the first stroke end, and the second direction refers to a direction facing the second stroke end.

(108) As shown in FIGS. 23-26, the present invention further provides a connection structure, which achieves a simple connection between the jump buckle **60** and the bracket **50**, to be specific: as shown in FIG. 23, the connection structure includes a bracket **50**, a jump buckle **60** and a jump buckle shaft **11**; the bracket **50** includes a bracket connecting plate **502**, and bracket arms **501** connected to the bracket connecting plate **502**; the jump buckle shaft **11** is connected to the bracket arms **501**; the jump buckle **60** is rotatably disposed on the jump buckle shaft **11**; and the bracket **50** further includes jump buckle positioning arms **503**, which limit two sides of the jump buckle **60**, thereby limiting a position of the jump buckle **60** in an axial direction of the jump buckle shaft **11**. The connection structure, compared with the prior art mode in which the jump buckle **60** and the jump buckle shaft **11** are riveted and then assembled onto the bracket **50**, is simpler and convenient to operate, and also reduces the requirements for a thermal treatment process of the jump buckle **11**.

(109) Further, as shown in FIGS. 24 and 26, a spacing W_0 between parts of the jump buckle positioning arms **503**, which are located on two sides of the jump buckle **60** and used for limiting the jump buckle **60**, is matched with the thickness of the jump buckle **60**, ensuring the rotation flexibility of the jump buckle **60** while preventing the jump buckle **60** from moving along an extension direction of the jump buckle shaft **11**.

(110) Further, as shown in FIGS. 23-24, and 26, at least two jump buckle positioning arms **503** are disposed to face each other at intervals. Further, as shown in FIGS. 23 and 24, the two jump buckle positioning arms **503** are staggered axially along the jump buckle shaft **11**, and are located on two sides of the jump buckle shaft **11**.

(111) Further, as shown in FIGS. 23-24 and 26, an implementation of the jump buckle positioning arms **503** is as follows: two jump buckle positioning arms **503** are disposed to face each other at intervals, wherein one end of each jump buckle positioning arm **503** is connected to the bracket connecting plate **502**, and the other end of the jump buckle positioning arm **503** is blocked on one side of the jump buckle **60**. Further, as shown in FIG. 23, one end of each of the two jump buckle

positioning arms **503** is respectively connected to the bracket connecting plate **502** by bending, the other end of the jump buckle positioning arm **503** extends in the direction where the jump buckle shaft **11** is located and is respectively blocked on two sides of the jump buckle **60**, and the length of the jump buckle positioning arm **503** is greater than a spacing between the jump buckle shaft **11** and the bracket connecting plate **502**. Specifically, the side facing a reader in FIG. **23** is a front side, the rear end of the jump buckle positioning arm **503** is connected to the bracket connecting plate **502**, and the front end of the jump buckle positioning arm **503** extends in the direction where the jump buckle shaft **11** is located. Further, as shown in FIG. **23**, the jump buckle positioning shaft **503** and the bracket connecting plate **502** are of an integrated structure, which is formed by cutting and bending the middle part of the bracket connecting plate **502**.

(112) Further, the following is another implementation of the jump buckle positioning arm **503** (not shown): the bracket **50** further includes a positioning arm connecting plate, wherein one end of the positioning arm connecting plate is connected to two jump buckle positioning arms **503**, and the other end of the positioning arm connecting plate is connected to the bracket connecting plate **502**.

(113) It should be pointed out that a setting method of the jump buckle positioning arms **503** is not limited to the above two implementations. The jump buckle positioning arms **503** may also be connected to the bracket arms **501**; or the jump buckle positioning arms **503** and the bracket **50** may be of an integrated structure, or of split structures assembled together later (by common connection means, such as welding, screw connection and riveting).

(114) Further, each jump buckle positioning arm **503** includes a positioning arm avoidance hole for the jump buckle shaft **11** to pass through; or, as shown in FIG. **23**, the jump buckle positioning arm **503** includes a semicircular positioning arm avoidance groove for the jump buckle shaft **11** to pass through, and opening ends of the two positioning arm avoidance grooves are opposite each other.

(115) As shown in FIGS. **23** and **24**, an embodiment of the jump buckle positioning arms **503** is as follows: two jump buckle positioning arms **503** are staggered in an axial direction of the jump buckle shaft **11** and are located on two sides of the jump buckle shaft **11**; and as shown in FIG. **23**, the jump buckle positioning arm **503** includes a semicircular positioning arm avoidance groove for the jump buckle shaft **11** to pass through, and opening ends of the two positioning arm avoidance grooves are opposite each other. as shown in FIGS. **23-24**, and **26**, an embodiment of the connection structure is as follows: as shown in FIGS. **23** and **24**, the bracket **50** is of a U-shaped structure, including a bracket connecting plate **502**, and two bracket arms **501** connected to two ends of the bracket connecting plate **502** respectively by bending; as shown in FIG. **26**, two ends of the jump buckle shaft **11** are connected to the two bracket arms **501** respectively; as shown in FIGS. **23**, **24** and **26**, the jump buckle **60** is rotatably disposed on the jump buckle shaft **11**; the bracket **50** further includes two jump buckle positioning arms **503** located between the two bracket arms **501** and disposed to face each other at intervals, wherein the two jump buckle positioning arms **503** are disposed on two sides of the jump buckle **60** to block the jump buckle **60**, thereby limiting the range of movement of the jump buckle **60** along an extension direction of the jump buckle shaft **11** (that is, limiting the position of the jump buckle **60** in the axial direction of the jump buckle shaft **11**).

(116) As shown in FIGS. **17-21**, a fifth embodiment of the operating mechanism **100** is specifically as follows: as shown in FIGS. **17-21**, the operating mechanism **100** includes a bracket **50**, a rocker arm assembly and a jump buckle **60** disposed on the bracket **50** respectively, a first crank **30** pivotally disposed on the jump buckle **60** around the first axis **67m**, and a first spring **22**; one end of the jump buckle **60** is rotatably connected to the bracket **50**, and is a jump buckle pivoting end; the jump buckle **60** includes a jump buckle hole **605** for a positioning pin shaft **17** to be inserted, the bracket **50** includes a bracket hole **508** for the positioning pin shaft **17** to be inserted, the jump buckle hole **605** and the bracket hole **508** are aligned, and the first spring shaft **46** is in limiting fit with the jump buckle **60**; one end of the first crank **30** swings away from the jump buckle pivoting end, so that a spacing between the first spring shaft **46** and the second spring shaft **16** is less than or

equal to a length of the first spring **22**, forming a first assembly state; and in the first assembly state, two ends of the first spring **22** are assembled to the first spring shaft **46** and the second spring shaft **16** respectively, and the rocker arm assembly swings in a direction where the jump buckle pivoting end is located and drives the first spring **22** and the first crank **30** to swing to the jump buckle pivoting end respectively, thereby forming a second assembly state. According to the operating mechanism **100** in this embodiment, the first spring **22** may be mounted on the first spring shaft **46** and the second spring shaft **16** simply and rapidly, thereby improving the assembly efficiency of the operating mechanism **100** and saving the assembly time and labor cost.

(117) Specifically, as shown in FIG. **17**, the jump buckle hole **605** is aligned with the bracket hole **508**; the first spring shaft **46** is in limiting fit with the jump buckle **60**, and one end of the first crank **30** swings in a direction away from the jump buckle pivoting end till being in limiting fit with the jump buckle **60**, such that the first assembly state is formed. Further, as shown in FIGS. **17** and **18**, in the first assembly state, two ends of the first spring **22** are assembled to the first spring shaft **46** and the second spring shaft **16**, respectively; and after the rocker arm assembly swings so that the first spring shaft **46** moves away from the jump buckle **60** and the axis line of the first spring **22** swings over the first axis **67m**, the first spring **22** drives the rocker arm assembly to swing to one end of its swing stroke, while driving the first crank **30** to swing in the direction of the jump buckle pivoting end to the first crank **30** till being limiting fit with the jump buckle **60** again. At this time, the assembly of the first spring **22** is completed, and the operating mechanism **100** enters the second assembly state as shown in FIG. **20**. Further, in a direction shown in FIGS. **17-18**, and **20-21**, the right end of the jump buckle **60** is the jump buckle pivoting end; “one end of the first crank **30** swings away from the jump buckle pivoting end” refers that the lower end of the first crank **30** swings clockwise; and “the first crank **30** swings in the direction of the jump buckle pivoting end” refers that the lower end of the first crank **30** swings counterclockwise.

(118) Further, as shown in FIGS. **17** and **25**, the jump buckle **60** further includes a jump buckle protrusion **66**; and in the first assembly state, the jump buckle protrusion **66** limits a swing position of the first crank **30**. Further, as shown in FIG. **17**, the jump buckle protrusion **66** in the first assembly state is in limiting fit with the first crank **30**. Further, as shown in FIGS. **17** and **18**, the first spring shaft **46** and the second spring shaft **16** are located on two sides of the jump buckle **60**; and the jump buckle protrusion **66** is located between the jump buckle shaft **11** and the jump buckle hole **605**, and the first axis **67m** is located between the jump buckle protrusion **66** and the jump buckle pivoting end. Specifically, in the direction shown in FIGS. **17** and **18**, the first spring shaft **46** and the second spring shaft **16** are located on the upper and lower sides of the jump buckle **60**, respectively.

(119) Further, as shown in FIG. **17** and FIG. **18**, the rocker arm assembly is disposed in the V-shaped groove of the bracket **50**; the jump buckle pivoting end is located on one side of the V-shaped groove, and the latch **13**, the re-buckle **15** and the bracket hole **508** are located on the other side of the V-shaped groove; and the rocker arm **45** is pivotally disposed at the bottom of the V-shaped groove. Specifically, in the direction shown in FIG. **17** and FIG. **18**, the jump buckle pivoting end is located on the right side of the V-shaped groove, and the latch **13**, the re-buckle **15** and the bracket hole **508** are located on the left side of the V-shaped groove.

(120) As shown in FIG. **25**, an embodiment of the jump buckle **60** is as follows: the jump buckle **60** is of a strip plate structure, wherein a jump buckle shaft hole **601** and a limiting shoulder **602** which is in limiting fit with the bracket connecting plate **502** of the bracket **50** are formed at one end of the jump buckle **60**, a jump buckle hole **605** and a jump buckle table **604** which is in latching fit with the latch **13** are formed in the other end of the jump buckle **60**, and the jump buckle protrusion **66** and a jump buckle-crank shaft hole are formed in the middle of the jump buckle **60**; the jump buckle hole **605**, the jump buckle protrusion **66**, the jump buckle-crank shaft hole and the jump buckle shaft hole **601** are arranged side by side at intervals in sequence; and two edges of the jump buckle **60** in a length direction are provided with a driving side edge **603** and a limiting side edge

608, respectively, which are located at two ends of the jump buckle **60** in the length direction.

(121) Based on the fifth embodiment of the operating mechanism **100**, the present invention further provides an assembly method of an operating mechanism, which can complete the assembly of the first spring **220** easily and quickly and is conducive to improving the assembly efficiency of the entire operating mechanism **100** and realizing automatic assembly. The assembly method of the operating mechanism includes the following steps.

(122) Step I, Aligning the jump buckle hole **605** of the jump buckle **60** to the bracket hole **508** of the bracket **50**, and inserting the positioning pin shaft **17** inside the jump buckle **605** and the bracket hole **508**, such that the operating mechanism enters the first assembly state.

(123) Further, in the step I, the jump buckle hole **605** is aligned with the bracket hole **508**, the positioning pin shaft **17** is mounted in the jump buckle hole **605** and the bracket hole **508**, and the rocker arm **45** swings away from the jump buckle pivoting end, such that the first spring shaft **46** is in limiting fit with the jump buckle **60**; and the first crank **30** swings away from the jump buckle pivoting end to be in limiting fit with the jump buckle **60**, such that the operating mechanism **100** enters the first assembly state. At this time, a spacing between an axis line of the first spring shaft **46** and an axis line of the second spring shaft **16** is less than or equal to the length of the first spring **22**.

(124) Step II, Assembling two ends of the first spring **22** to the first spring shaft **46** and the second spring shaft **16** respectively in the first assembly state; and swinging the rocker arm **45** toward the jump buckle pivoting end, wherein the rocker arm **45** drives the first spring **22** and the first crank **30** to rotate, such that the operating mechanism enters the second assembly state.

(125) Further, in the step II, two ends of the first spring **22** are assembled to the first spring shaft **46** and the second spring shaft **16**, respectively; the rocker arm **45** swings in the direction where the jump buckle pivoting end is located, the rocker arm **45** drives the first spring **22** through the first spring shaft **46** to swing around the second spring shaft **16**, and the axis of the first spring **22** swings over the rotation center (i.e., the first axis **67m**) of the first crank **30**; the first spring **22** drives the rocker arm **45** to swing to one end of the swing stroke of the rocker arm assembly, and meanwhile, the first spring **22** drives the first crank **30** to swing in the direction of the jump buckle pivoting end to the first crank **30** till being in limiting fit with the jump buckle **60** again. The operating mechanism enters the second assembly state, and the first spring **22** completes the assembly.

(126) Further, the assembly method of the operating mechanism of the present invention further includes a step III, i.e., assembling the reset structure **42** of the rocker arm assembly on the rocker arm **45** in the second assembly state, drawing the positioning pin shaft **17** off, and driving the jump buckle **60** by the first spring **22** to rotate to be in limiting fit with the reset structure **42**.

(127) Further, the assembly method of the operating mechanism of the present invention further includes steps IV and V, the orders of which may be interchangeable: in the step IV, the slider **26** is assembled on the slide rails **25**, and two ends of the first connecting rod **27** are rotatably assembled on the second spring shaft **16** and the slider **26**, respectively. In the step V, the latch **13** is pivotally disposed on the bracket **50** through the latch shaft **12** respectively, and the re-buckle **15** is pivotally disposed on the bracket **50** through a re-buckle shaft **14**.

(128) Further, the assembly method of the operating mechanism of the present invention further includes the following operations: prior to the step I, the second spring shaft **16** is assembled on the first crank **30**, the first crank **30** is pivotally disposed on the jump buckle **60** around the first axis **67m**, and the jump buckle **60** is pivotally disposed on the bracket **50**; and the first spring shaft **46** is mounted on the rocker arm **45** of the rocker arm assembly, and the rocker arm **45** is pivotally disposed in the V-shaped groove of the bracket **50**.

(129) Further, as shown in FIGS. **7-16**, and **35**, the moving contact mechanism further includes a contact spring **23**, wherein one end of the contact spring **23** is connected to the moving contact **9**, and the other end of the contact spring **23** is connected to the contact support **110**. When the

moving contact **9** and the static contact **18** are closed, a first acting force is applied to the moving contact **9** so that the moving contact **9** presses the static contact **18**. Further, as shown in FIGS. 7-16, and 35, one end of the contact spring **23** is connected to the moving contact **9** through a third spring shaft **201**, and the other end of the contact spring **23** is rotatably connected to the contact support **110** through the fourth spring shaft **202**. Further, as shown in FIGS. 29 and 33, the moving contact **9** includes a moving conductive rod **90**, wherein the moving conductive rod **90** is provided with a conductive rod clamping groove **902** cooperated with the third spring shaft **201**.

(130) Further, as shown in FIGS. 14-16, the contact spring **23** may also achieve the locking of the moving contact **9**, to be specific: two ends of the contact spring **23** are a third spring end and a fourth spring end respectively, wherein the third spring end is connected to the moving contact **9**, the fourth spring end is connected to the contact support **110**, a geometric axis line of the contact spring **23** is a second axis line, and the second axis line coincides with a connecting line of the third spring end and the fourth spring end; as shown in FIG. 11, the second axis line is located on one side of the third axis **111s** when the moving contact **9** is normally closed or disconnected, and the contact spring **23** keeps the moving contact **9** in a normally closed position or a normally disconnected position; the moving contact **9** rotates relative to the contact support **110** when the moving contact **9** is repelled by an electric repulsion force generated by a short-circuit current; and the moving contact **9** drives the contact spring **23** to rotate around the fourth spring end, so that the second axis line swings to the other side of the third axis **111s**, and the moving contact **9** remains in a temporary breaking position. The moving contact mechanism includes a contact support **110**, a moving contact **9** and a contact spring **23**. The moving contact mechanism has a simple structure. The contact spring **23** realizes the overtravel of the moving contact **9** to ensure the reliable contact between the moving contact **9** and the static contact **18**; and the contact spring **23** locks the moving contact **9** in the temporary breaking position when the moving contact **9** is repelled by an electric repulsion force generated by the short-circuit current, so that the moving contact **9** does not rebound in the occurrence of a short-circuit fault, which ensures the reliable breaking of the moving contact **9** and the static contact **18**. It should be pointed out that when the moving contact **9** is located in the temporary breaking position, if the operating mechanism **100** switches from the closed state to the disconnected state, the moving contact **9** automatically acts from the temporary breaking position to a normal disconnecting position.

(131) Further, as shown in FIG. 11, the moving contact **9** and the static contact **18** are closed, and the short-circuit current flows through the moving contact **9** and the static contact **18**. Due to a current direction in the moving contact **9** being opposite to a current direction of a part, opposite to the moving contact **19**, of the static contact **18**, an electric repulsion force is generated between the moving contact **9** and the static contact **18**, so that the moving contact **9** is repulsed.

(132) Further, as shown in FIGS. 14-16, the contact spring **23** passes through a second dead center position when the moving contact **9** drives the contact spring **23** to rotate and thus the second axis line swings from one side of the third axis **111s** to its other side; and as shown in FIG. 15, the third axis **111s** is located on the second axis line when the contact spring **23** is located in the second dead center position.

(133) Specifically, as shown in FIGS. 11 and 12, in the case that the circuit breaker of the present invention is normally closed or normally disconnected, the contact spring **23** and the contact support **110** act synchronously and are relatively stationary. The second axis line of the contact spring **23** is always kept on the same side of the third axis **111s**, and undergoes a small deformation only when the moving contact **9** and the static contact **18** are closed, so as to provide an overtravel force for the moving contact **9**, thereby ensuring that the moving contact **9** and the static contact **18** are tightly closed. In the direction shown in FIGS. 14-16, when the short-circuit current flows through the circuit breaker of the present invention, the moving contact **9** will be repelled by the electric repulsion force generated by the short-circuit current, so that the moving contact **9** rotates in a counterclockwise direction relative to the contact support **110** (the contact support **110** remains

stationary because the operating mechanism is in the closed state), and the moving contact **9** (through the third spring shaft **201**) drives the contact spring **23** to rotate counterclockwise around the fourth spring end. As shown in FIG. **15**, when the contact spring **23** rotates to the second dead center position, the energy storage of the contact spring **23** reaches a maximum value, and the third axis **111s** is located on the second axis line. As shown in FIG. **16**, the second axis line also passes the third axis **111s** while the contact spring **23** passes the second dead center position. Therefore, the third axis **111s** may also be regarded as the second dead center position, that is, the second axis passes the third axis **111s**, i.e., the contact spring **23** passes the second dead center position. After the contact spring **23** passes the second dead center position, the energy is released and drives the moving contact **9** to quickly rotate to the temporary breaking position, so that the moving contact **9** remains in the temporary breaking position, and finally the second axis line is moved from the lower side of the third axis **111s** to its upper side.

(134) The present invention further discloses a moving contact assembly that can significantly improve the reliability of the connection between the conductor **70** and the moving contact **9**, and realize a hard connection therebetween, to be specific: as shown in FIGS. **29-34**, and **45-46**, the moving contact assembly includes a conductor **70**, an inelastic fastener **80** and a moving contact **9**, wherein the conductor **70** includes a first clamping arm **710** and a second clamping arm **711** which are disposed to face each other at intervals; the moving contact **9** includes a moving conductive rod **90** and a moving contact point **94**; the moving contact point **94** is disposed at one end of the moving conductive rod **90**; the moving conductive rod **90** includes a conductive rod contact portion disposed at the other end; the conductive rod contact portion is inserted between the first clamping arm **710** and the second clamping arm **711** and rotatably connected to the first clamping arm **710** and the second clamping arm **711**, respectively; and the fastener **80** is connected to the first clamping arm **710** and the second clamping arm **711** respectively, so that the first clamping arm **710** and the second clamping arm **711** clamp the conductive rod contact portion. Further, as shown in FIGS. **29-34**, and **45-46**, the conductor **70** further includes a conductor connecting plate **712**, and two ends of the conductor connecting plate **712** are connected to the first clamping arm **710** and the second clamping arm **711** respectively by bending.

(135) Compared with the prior art, such as in the Japanese patent JP3794163B2 in which a double-torsion-spring pressing conductor and a moving contact are adopted, the moving contact assembly of the present invention is characterized in that the fastener **80** achieves a hard connection between the conductor **70** and the conductive rod **90**, which, under the premise of ensuring that the moving conductive rod **90** has a certain flexibility of action, ensures the reliable structure and electrical connection between the conductor **70** and the conductive rod contact portion.

(136) It should be pointed out that the “inelastic fastener **80**” means that the fastener **80** will not undergo an elastic deformation caused by external forces.

(137) Further, as shown in FIGS. **30, 31, 34** and **45**, the conductor **70** further includes a conductor connecting plate **712**, and two ends of the conductor connecting plate **712** are connected to the first clamping arm **710** and the second clamping arm **711** by bending, respectively; and as shown in FIGS. **29** and **30**, the fastener **80** is disposed between the conductor connecting plate **712** and the conductive rod contact portion, so that the first clamping arm **711** and the second clamping arm **710** clamp the conductive rod contact portion. Further, as shown in FIGS. **30, 31, 34**, and **45**, the conductor connecting plate **712**, the first clamping arm **710** and the second clamping arm **711** are of a U-shaped structure as a whole. It should be pointed out that a clamping force of the first clamping arm **710** and the second clamping arm **711** for the moving conductive rod **90** may be adjusted by changing a length of a rivet body **802** and/or a position of the fastener **80** between the conductor connecting plate **712** and the moving conductive rod **90**.

(138) Further, as shown in FIGS. **30, 31, 34** and **45**, the first clamping arm **710** and the second clamping arm **711** each include a clamping arm straight portion and a clamping arm bent portion, wherein two ends of the clamping arm bent portion are connected to the clamping arm straight

portion and the conductor connecting plate **712** respectively by bending; the two clamping arm bent portions are offset by the clamping arm straight portions of the first clamping arm **710** and the second clamping arm **711** to the middle of the clamping arm connecting plate **712**, respectively; the fastener **80** is disposed on the two clamping arm straight portions and fixedly connected to the two clamping arm straight portions, respectively; and the moving conductive rod **90** is rotatably connected to the two clamping arm straight portions. Further, as shown in FIGS. **31** and **34**, the clamping arm straight portion of the first clamping arm **710** is provided with a first clamping arm hole **7101** and a first clamping arm shaft hole **7102** (or a first clamping arm shaft table **7103**); and the clamping arm straight portion of the second clamping arm **711** is provided with a second clamping arm hole **7111** and a second clamping arm shaft hole **7112** (or a second clamping arm shaft table **7113**).

(139) Further, as shown in FIG. **38**, the unit housing **120** includes a wiring board slot **120-4** that is in plug-in fit with a conductor wiring board **700**.

(140) Further, as shown in FIG. **31**, in a state where the fastener **80** is not mounted, a spacing between the first clamping arm **710** and the second clamping arm **711** is $D_{sub.1}$. As shown in FIG. **30**, the thickness of the conductive rod contact portion is, $D_{sub.1} \geq D_{sub.0}$. The “in the state where the fastener **80** is not mounted” means that the first clamping arm **710** and the second clamping arm **711** are in an initial free state, but are not constrained by the fastener **80** if the fastener **80** and the conductor **70** have not been assembled together.

(141) Further, as shown in FIGS. **30** and **45**, an inner sidewall of the first clamping arm **710** is in point contact or line contact with the conductive rod contact portion, and an inner sidewall of the second clamping arm **711** is in surface contact with the conductive rod contact portion. A contact mode in which the first clamping arm **710** and the second clamping arm **711** are in contact with the conductive rod contact portion is conducive to increasing a contact area between the conductor **70** and the moving contact **9**, improving the conductivity of the moving contact mechanism, and maintaining the movement performance between the conductor and the moving contact.

(142) Further, as shown in FIGS. **30** and **45**, one end of the fastener **80** is fixedly connected or in clamping connection with the first clamping arm **710**, and the other end of the fastener **80** is fixedly connected to the second clamping arm **711**. Further, the fastener **80** is a rivet, wherein one end of the rivet is a rivet head **801** in clamping connection with the first clamping arm **710**, and the other end of the rivet is a riveting end **803** fixedly connected to the second clamping arm **711**.

(143) Further, as shown in FIG. **32**, the fastener **80** is a rivet. Further, as shown in FIG. **32**, an embodiment of the fastener **89** is as follows: the fastener **80** includes a rivet head **801**, a rivet body **802** and a rivet end **803** which are disposed in sequence, wherein an outer diameter of the rivet head **801** is greater than an outer diameter of the rivet body **802**, and a first annular table **804** is disposed at the connection between the rivet head **801** and the rivet body **802**; and an outer diameter of the rivet body **802** is greater than an outer diameter of the rivet end **803**, and a second annular table **805** is disposed at the connection between the rivet body **802** and the rivet end **803**. As shown in FIGS. **30** and **45**, the first annular table **804** is in limiting fit with the first clamping arm **710**, and the rivet body **802** passes through the first clamping arm **710** so that the second annular table **805** is in surface contact with the second clamping arm **711**. The thickness of the first clamping arm **710** is, and the length of the rivet body (**802**) is, $L_{sub.0} < D_{sub.1} + D_{sub.3}$.

(144) It should be pointed out, as shown in FIGS. **30** and **45**, the inner sidewall of the first clamping arm **710** is in line contact or point contact with the conductive rod contact portion, and the second clamping arm **711** is in surface contact with the conductive rod contact portion. The reasons for the above contact modes are as follows: a second annular table **805** is in surface contact with the second clamping arm **711** in the course of riving the rivet, so the rivet head **801** will deform the connection (a bend **720**) between the first clamping arm **710** and the conductor connecting plate **712**, such that the first clamping arm **710** inclines in a direction where the second clamping arm **711** is located. Therefore, the first clamping arm **710** is in line contact or point

contact with the moving conductive rod **90**, and the moving conductive rod **90** is in surface contact with the second clamping arm **711**. Therefore, a contact area between the conductive rod contact portion and the conductor **70** is significantly increased, the conductivity of the moving contact mechanism is improved, and the heat generation in the conductive process of the moving contact assembly is reduced to prolong the service life of the moving contact assembly.

(145) Further, as shown in FIGS. **31** and **34**, the first clamping arm **710** is provided with a first clamping arm hole **7101** for the rivet body **802** to pass through, the second clamping arm **711** is provided with a second clamping arm hole **7111** for the rivet end **803** to pass through, and an inner diameter of the first clamping arm hole **7101** is greater than an inner diameter of the second clamping arm hole **7111**.

(146) As shown in FIG. **31**, the first connection mode of the moving conductive rod **90** and the conductor **70** is as follows: the first clamping arm **710** is further provided with a first clamping arm shaft hole **7102**, and the second clamping arm **711** is provided with a second clamping arm shaft hole **7112**. As shown in FIG. **30**, the moving contact assembly further includes a contact shaft **10**, two ends of which are inserted in the first clamping arm shaft hole **7102** and the second clamping arm shaft hole **7112** respectively; the moving conductive rod **90** is of a strip-shaped plate structure, one end of which is the conductive rod contact portion; and the conductive rod contact portion is rotatably disposed on the contact shaft **10**. Further, the moving contact mechanism includes two contact springs **23**, which are disposed on both sides of the moving contact **9** respectively, wherein one end of each contact spring **23** is connected to the moving contact **9** through a third spring shaft **201**, and the other end of the contact spring **23** is connected to the contact support **110** through the fourth spring shaft **202**.

(147) As shown in FIG. **34**, the second connection mode of the conductive rod **90** and the conductor **70** is as follows: the first clamping arm **710** is further provided with a first clamping arm shaft table **7103**, and the second clamping arm **711** is further provided with a second clamping arm shaft table **7113**; the moving conductive rod **90** is of a strip plate structure, wherein a conductive rod contact portion is disposed at one end of the moving conductive rod **90**; the conductive rod contact portion is provided with a conductive rod shaft hole **901**; and the first clamping arm shaft table **7103** and the second clamping arm shaft table **7113** are inserted opposite to each other in the conductive rod shaft hole **901**. Further, as shown in FIG. **34**, the first clamping arm shaft table **7103** and the second clamping arm shaft table **7113** are respectively annular tables, which are formed by stamping the first clamping arm **710** and the second clamping arm **711** opposite to each other. The first clamping arm shaft table **7103** and the second clamping arm shaft table **7113** are conducive to increasing a contact area between the conductor **70** and the moving conductive rod **90**. Further, the moving contact mechanism includes two contact springs **23**, which are disposed on both sides of the moving contact **9** respectively, wherein one end of each contact spring **23** is connected to the moving contact **9** through a third spring shaft **201**, and the other end of the contact spring **23** is connected to the contact support **110** through a fourth spring shaft **202**.

(148) As shown in FIG. **45** and FIG. **46**, the third connection mode of the conductive rod **90** and the conductor **70** is as follows: the moving conductive rod **90** further includes a conductive rod body **90-2**, wherein a moving contact point **94** is disposed at one end of the conductive rod body **90-2**, and the other end of the conductive rod body **90-2** is connected to the conductive rod contact portion; the conductive rod contact portion includes a contact portion bottom plate, and two conductive rod contact plates **907** which are connected to two ends of the contact portion bottom plate respectively by bending and disposed to face each other at intervals; and the first clamping arm **710** and the second clamping arm **711** are rotatably connected to the two conductive rod contact plates **907** through the contact shaft **10**, respectively. Further, as shown in FIGS. **45** and **46**, the conductive rod contact portion is of a U-shaped structure, in which the conductive rod body **90-2** is connected to the middle part of one side of the contact portion bottom plate and the conductive rod contact plates **907** are located on two sides of the contact portion bottom plate, respectively.

The moving contact mechanism includes at least one contact spring **23**, wherein the contact spring **23** is located between the two conductive rod contact plates **907**, one end of the contact spring **23** is connected to the moving contact **9** through the third spring shaft **201**, and the other end of the contact spring **23** is connected to the contact support **110** through the fourth spring shaft **202**. Further, as shown in FIGS. **46**, two conductive rod contact plates **907** is respectively provided with a moving contact clamping groove **902** cooperated with the third spring shaft **203**.

(149) Further, as shown in FIGS. **11-16**, **35**, **37-40**, and **45**, the moving contact mechanism further includes a moving contact insulator **140**, wherein the cooperation between the moving contact insulator **140** and the moving contact **9** can significantly increase an insulation gap and a creepage distance between the moving contact **9** and the static contact **18**. In addition, the moving contact insulator **140** can prevent electric arc particles generated when the moving contact **9** and the static contact **18** are broken from entering into the contact support **110** and from being attached to the contact spring **23** to affect its elasticity, and from being attached to a rotating shaft of the moving contact **9** to affect its action performance. The following is an implementation of the moving contact insulator **140**, to be specific: the moving contact insulator **140** includes an insulator body which includes an insulator bottom plate **140-9** and an insulator sidewall **140-1**; a moving contact accommodating cavity **140-2** used to accommodate the moving contact **9** is formed in the middle of the insulator body; the moving contact insulator further includes a main baffle **140-4** and a main isolation plate **140-5**, wherein the main baffle **140-4** is disposed outside the insulator bottom plate **140-9** and extends downward, the main isolation plate **140-5** is vertically connected to the insulator sidewall **140-4** and protrudes out of the insulator sidewall **140-4**, and the main isolation plate **140-5** extends in a length direction of the insulator body. The moving contact insulator **140** is simple in structure and convenient to assemble, and can significantly improve the insulation performance of the moving contact **9**, and increase the creepage distance between the moving contact **9** and the static contact **18**. Further, as shown in FIGS. **35** and **37**, the main isolation plate **140-5** extends from one end of the insulator body to the other end of the insulator body. Specifically, as shown in FIGS. **35** and **37**, one end of the insulator body is close to the moving contact point of the moving contact **9**, and is a first body end; the other end of the insulator body is a second body end; and the main isolation plate **140-5** extends from the first body end to the second body end.

(150) Further, as shown in FIG. **37**, the main baffle **140-4** and the main isolation plate **140-5** are of an integrated structure. Further, as shown in FIG. **37**, the main baffle **140-4**, the main isolation plate **140-5** and the insulator body are of an integrated structure.

(151) Further, as shown in FIG. **37**, two insulator sidewalls **140-1** are disposed to face each other at intervals; the main isolation plate **140-5** is respectively disposed on two sides of the insulator body respectively; the two main isolation plates **140-5** are disposed vertically on two sides of the two insulator sidewalls **140-4** respectively and protrude to two sides of the two insulator sidewalls **140-1** respectively; and one end of each main isolation plate **140-5** is connected to one end of one main baffle **140-4**. Further, as shown in FIG. **37**, a cross section of the insulator body is of a U-shaped structure, and the main baffle **140-4** and the main isolation plate **140-5** are of a U-shaped structure as a whole, which encloses the outside of the insulator body.

(152) Specifically, in a direction shown in FIG. **37**, the moving contact accommodating cavity **140-2** is located on the upper side of the insulator bottom plate **140-9** (also the inner side of the insulator bottom plate **140-9**); the two main isolation plates **140-5** are vertically connected to the left and right sides of the two insulator sidewalls **140-1**, respectively; the main baffle **140-4** is disposed on the lower side of the insulator bottom plate **140-9** (also the outer side of the insulator bottom plate **140-9**) and extends to the lower side of the insulator bottom plate **140-9**; and the main isolation plate **140-5** extends from the front end of the insulator body to the rear end of the insulator body (i.e., both ends of the insulator body in a length direction).

(153) Further, as shown in FIGS. **35** and **37**, the main isolation plate **140-5** is formed into a flared mouth shape as a whole, wherein an opening direction of the flared mouth shape is oriented

towards the outer side of the insulator bottom plate **140-9**. Further, an opening direction of the flared mouth shape of the main isolation plate **140-5** is opposite to an opening direction of the moving contact accommodating cavity **140-2**. Specifically, in the direction shown in FIG. 37, the opening direction of the flared mouth shape faces downward, and the opening direction of the moving contact accommodating cavity **140-2** faces upward.

(154) Further, as shown in FIGS. 35 and 37, the main isolation plate **140-5** includes an isolation plate head **140-50**, an isolation plate neck **140-51**, an isolation plate abdomen **140-52** and an isolation plate tail **140-53**, which are connected in sequence, wherein one end of the isolation plate tail **140-53** is connected to the main baffle **140-4**; and the isolation plate head **140-50** is disposed near the moving contact point **94** of the moving contact **9**. Further, as shown in FIGS. 35 and 37, the upper side of the isolation plate abdomen **140-52** is flush with an opening side of the moving contact accommodating cavity **140-2**.

(155) As shown in FIGS. 14-16, and 35-37, an embodiment in which the movable contact **9** and the moving contact insulator **140** cooperate with the contact support **110** is as follows: the moving conductive rod **90** of the moving contact **9** is inserted in the moving contact accommodating cavity **140-2**, and the moving contact **9** and the moving contact insulator **140** form a first assembly; a support assembling cavity **110-0** is formed in the middle of the contact support **110**, the first assembly and the contact spring **23** are respectively disposed in the support assembling cavity **110-0**, and the insulator bottom plate **140-9** props against a support bottom wall **110-9** of the support assembling cavity **110-0**; the moving contact insulator **140** protrudes at one side of the contact support **110** from one end of the support assembling cavity **110-9**; and the moving contact **9** drives the moving contact insulator **140** to rotate synchronously while the moving contact **9** rotates relative to the contact support **110** under an electric repulsion force generated by a short-circuit current, such that an exposure gap is formed between the insulator bottom plate **140-9** and a bottom wall of the support assembling cavity **110-9**, and the main baffle **140-4** blocks the exposure gap on one side of the contact support **110**.

(156) Specifically, in a direction shown in FIG. 14, a huge electric repulsion force causes the moving contact **9** to be repulsed and rotated counterclockwise while the short-circuit current flows through the circuit breaker of the present invention, such that an exposure gap having an angle of θ is formed between the insulator bottom plate **140-9** and the support bottom wall **110-9**. When the moving contact **9** is initially separated from the static contact **18**, a large number of electric arc particles will be generated. However, the main baffle **140-4** is located on the right side of the contact support **110** to block the exposure gap, thereby preventing electric arc particles from entering the contact support assembling cavity through the exposure gap, and from being deposited in the contact spring **23** and/or contact shaft **10** to affect the operation performance of the moving contact mechanism.

(157) Further, as shown in FIG. 37, the moving contact insulator further includes secondary baffles **140-7** which are respectively disposed on both sides of the insulator body; the secondary baffles **140-7** and the main baffle **140-5** are arranged side by side at intervals, and the secondary baffles **140-7** and the main isolation plates **140-5** are located on both sides of the main baffle **140-4**, respectively; the secondary baffles **140-7** are vertically connected to the insulator sidewall **140-1** and protrude toward the outside of the insulator sidewall **140-1**; and one end of each of the secondary baffles **140-7** protrudes on one side of the insulator bottom plate **140-9** to form a secondary baffle protrusion, and the secondary baffle protrusions located on both sides of the insulator body are connected to each other. Further, as shown in FIG. 37, the two secondary baffles **140-7** are disposed vertically on both sides of the two insulator sidewalls **140-2** respectively, and protrude to both sides of the two insulator sidewalls **140-2** respectively, and one ends of the two secondary baffles **140-70** protrude on one side of the insulator bottom plate **140-9** and are connected to each other, so that the two secondary baffles **140-70** form a U-shaped structure as a whole.

(158) Specifically, in the direction shown in FIG. 37, the secondary baffles **140-7** are disposed at the rear end of the insulator body and are vertically connected to the left and right sides of the insulator sidewall **140-1**, respectively; the lower ends of the two secondary baffles **140-7** protrude from the lower side of the insulator bottom plate **140-9** respectively and are connected into a whole, and the secondary baffles **140-7** are located on the rear side of the main baffle **140-4** and are disposed side by side; and the upper end of each secondary baffle **140-7** is flush with the opening side of the moving contact accommodating cavity **140-2**.

(159) Further, as shown in FIG. 36, the secondary baffles **140-7** are located within the support assembling cavity **110-0**, respectively, and cooperate with the sidewall of the support assembling cavity **110-0** respectively, to block the exposure gap. Further, as shown in FIG. 37, the contact support **110** includes two support cooperating ribs disposed at one end of the support assembling cavity **110-0** and disposed to face each other at intervals, wherein the two support cooperating ribs are formed by bending inward from one ends of the two sidewalls of the support assembling cavity **110**, and the two support cooperating ribs are staggered and cooperated with the two secondary baffles **140-7** respectively, to block the exposure gap. The secondary baffles **140-7** cooperates with the contact support **110**, to further prevent the electric arc particles generated by the separation of the moving contact and the static contact from entering the support assembling cavity **110-0** through the exposure gap, which is conducive to prolonging the service life of the moving contact mechanism.

(160) Further, as shown in FIG. 35, the insulator body includes a first body section and a second body section which are connected by bending, wherein the main isolation plate **140-5** and the main baffle plate **140-4** are respectively connected to the first body section, and the secondary baffle **140-7** is connected to the second body section. Further, as shown in FIG. 35, the insulator body is of a <-shaped structure, and the shape of the insulator body is matched with the shape of the moving conductor rod **90** of the moving contact **9**.

(161) Further, as shown in FIG. 37, the insulator body includes a head connecting hole **140-3** and a tail connecting hole **140-6** which are formed in both ends thereof, respectively, for a head connecting pin and a tail connecting pin to be inserted respectively, such that the insulator body and the moving contact **9** are fixedly connected. Further, as shown in FIG. 35, in the case that the moving contact **9** and the moving contact insulator **140** are assembled, the moving conductive rod **90** of the moving contact **9** is inserted in the moving contact accommodating cavity **140-2**; as shown in FIG. 33, the moving conductive rod **90** includes a first moving contact connecting hole **905** and a second moving contact connecting hole **903** which are respectively formed in both ends thereof, wherein the first moving contact connecting hole **905** and the head connecting hole **140-3** are aligned, and the first connecting pin is inserted in the first moving contact connecting hole **905** and the head connecting hole **140-3**; and the second moving contact connecting hole **903** and the tail connecting hole **140-6** are aligned, and the tail connecting pin is inserted in the second moving contact connecting hole **903** and the tail connecting hole **140-6**, thereby achieving the fixed connection of the moving contact **9** and the moving contact insulator **140**. The moving contact **9** and the moving contact insulator **140** form the first assembly.

(162) As shown in FIG. 33, a first embodiment of the moving contact **9** is provided, wherein the moving contact **9** in this embodiment is a single-breakpoint moving contact: the moving contact **9** includes a moving conductive rod **90** and a moving contact point **94**, wherein the moving conductive rod **90** is of a strip plate structure, the moving contact point **94** is disposed at one end of the moving conductive rod **90**, and the conductive rod contact portion is disposed at the other end of the moving conductive rod **90**; the moving conductor rod **90** is provided with a first moving contact connecting hole **905**, a second moving contact connecting hole **903** and a moving contact shaft hole **901**, wherein the first moving contact connecting hole **905** and the moving contact shaft hole **901** are formed in both ends of the moving conductive rod **90**, respectively; the second moving contact connecting hole **903** is formed in the middle of the moving conductive rod **90** and

is provided near the moving contact shaft hole **901**; a moving contact clamping groove **902** is formed in the moving conductive rod **90**; and the conductive rod contact portion is provided with a contact protrusion **906**. Further, the moving conductive rod **90** is of a <-shaped structure, and the shape of the moving conductive rod is matched with the shape of the insulator body.

(163) Further, the conductive rod contact portion is of a circular plate structure, and the contact protrusion **906** which is in driving cooperation with the first push rod **150** is disposed on a circumferential sidewall of the conductive rod contact portion.

(164) It should be pointed out that the moving contact **9** in the first embodiment is suitable for a connection mode of the first and second moving contacts **9** and the conductor **70**.

(165) As shown in FIG. **46**, a second embodiment of the moving contact **9** is provided, wherein the moving contact **9** in this embodiment is a single-breakpoint moving contact: the moving contact **9** includes a moving conductive rod **90** and a moving contact point **94**, wherein the moving conductive rod **90** includes a conductive rod body **90-2** and a conductive rod contact portion, the moving contact point **94** is disposed at one end of the conductive rod body **90-2**, and the other end of the conductive rod body **90-2** is connected to the conductive rod contact portion; the conductive rod contact portion is of a U-shaped structure, including a contact portion bottom plate and two conductive rod contact plates **907** which are connected to both ends of the contact portion bottom plate by bending and disposed to face each other at intervals, respectively; the conductive rod body **90-2** is connected to the middle of one side of the contact portion bottom plate, and the conductive rod body **90-2** and the conductive rod contact plate **907** are located on both sides of the contact portion bottom plate, respectively; a first moving contact connecting hole **905** and a second moving contact connecting hole **903** (not shown) are formed in both ends of the conductor rod body **90-2**, respectively; and a moving contact clamping groove **902** is formed in one edge of one end, connected to the contact portion bottom plate, of the conductive rod contact plate **907**.

(166) It should be pointed out that the moving contact **9** in the second embodiment is suitable for the third connection mode of the moving contact **9** and the conductor rod **70**.

(167) As shown in FIG. **10**, a third embodiment of the moving contact is provided, wherein the moving contact **9** in this embodiment is a double-breakpoint moving contact: the moving contact **9** is of a central symmetrical structure, including a moving conductive rod **90** and two moving contact points **94** disposed at both ends of the moving conductive rod **90**, respectively, i.e., the first moving contact **94-0** and the second moving contact **94-1**, respectively, which cooperates with two static contacts **18** (the two static contacts **18** are a first static contact **18-0** and a second static contact **18-1**); and the moving contact **9** rotates, which can simultaneously realize the closure/disconnection with the two static contacts **18**. The moving contact **9** in this embodiment does not need to be electrically connected by the conductor **70**, but is directly disposed on the contact support **110**.

(168) As shown in FIG. **11** and FIG. **38**, an embodiment of the static contacts **18** is provided: the static contact **18** include a static contact bridge **18-1** and a static contact point **18-0** disposed at one end of the static contact bridge **18-1**; the static contact bridge **18-1** includes a U-shaped portion and a bending portion, wherein the bending portion is of a <-shaped structure, and the static contact **18-0** is disposed on one side arm of the U-shaped portion; the bending portion includes a first plate and a second plate connected by bending, wherein two ends of the first plate are connected to the U-shaped portion and the second plate by bending respectively, and the second plate is disposed parallel to the side arm of the U-shaped portion.

(169) As shown in FIG. **36**, an embodiment of the contact support **110** is provided: the contact support **110** is of a semi-cylindrical structure as a whole, including two support sidewalls **110-4** disposed to face each other at intervals, a support bottom wall **110-9** and a support assembling cavity **110-0**, wherein both ends of the support bottom wall **110-9** are connected to the two support sidewalls **110-4** respectively by bending, and the support assembling cavity **110-0** is formed between the two support sidewalls **110-4**; one ends of the two support sidewalls **110-4** are bent

inwardly to form two support cooperating ribs which are disposed to face each other at intervals, and two support clamping grooves **110-2** which cooperate with both ends of the fourth spring shaft **202** are formed at the inner sides of the other ends of the two support sidewalls **110-4**; and each support sidewall **110-4** is of a semicircular plate structure, wherein a support shaft groove **111** is formed in the circle center outside the support sidewall **110-4**, and a support connecting hole **110-5** is formed in the radial end of the support sidewall **110-4**.

(170) As shown in FIGS. **39-44**, the present invention also discloses a quick tripping device, in which the operating mechanism **100** quickly trips when the moving contact **9** is ejected due to a short-circuit fault of the circuit breaker, avoiding the moving contact **9** and the static contact **18** from being closed again. In addition, the operating mechanism cannot trip in a normal breaking/closing process of the moving contact **9** and the static contact **18**, and details are as follows.

(171) As shown in FIGS. **39-44**, the quick tripping device includes an operating mechanism, a moving contact mechanism and a static contact **18**, wherein the moving contact mechanism includes a contact support **110** and a moving contact **9**; the operating mechanism **100** is in driving connection to the moving contact mechanism, so that the moving contact **9** and the static contact **18** are closed or disconnected. The quick tripping device further includes a first push rod **150** pivotally disposed on the contact support **110**; the first push rod **150** includes a first push rod driven end and a first push rod driving end, wherein the first push rod driven end is in driving fit with the moving contact **9**, and the first push rod driving end cooperates with the operating mechanism **100** to trip the operating mechanism; and a driving gap is formed between the first push rod driven end and the moving contact **9**. When the moving contact **9** is electrically repelled by an electric repulsion force generated by a short-circuit current, the moving contact **9** rotates relative to the contact support **110**, and the moving contact **9** rotates around the driving gap, then contacts with the first push rod driven end and drives the first push rod **150** to rotate, so that the operating mechanism **100** is tripped. According to the quick tripping device of the present invention, the moving contact **9** and the contact support **110** rotate synchronously. Therefore, the driving gap between the first push rod driven end and the moving contact **9** remains unchanged while the moving contact mechanism rotates such that the moving contact **9** and the static contact **18** are closed or disconnected normally, but the moving contact **9** will rebound when the moving contact **9** is in contact with the static contact **18**. In the presence of the driving gap, a certain buffer space may be provided for the reasonable vibration generated when the moving contact **9** and the static contact **8** are closed, thereby avoiding the misoperation of the quick tripping device. However, if a short-circuiting fault occurs, the moving contact **9** is quickly repulsed by an electric repulsion force, and driven by the first push rod **150**, an intermediate transmission structure and the second push rod **18**, and the second push rod **18** drives the re-buckle **15** to release the limiting fit from the latch **13**, such that the latch **13** releases the latching fit from the jump buckle **60**, and the circuit breaker can be disconnected quickly.

(172) Specifically, as shown in FIG. **39**, the contact support **110** drives the first push rod **150** and the moving contact **9** to rotate clockwise/counterclockwise synchronously in response to the normal closing/disconnecting operation of the circuit breaker, such that a driving gap is always remained between the first push rod **150** and the moving contact **9**, and the quick tripping device will not be triggered. In particular, due to the hard contact between the moving contact **9** and the static contact **18** when the circuit breaker of the present invention is closed normally, the moving contact **9** will rebound to a certain extent. Due to the existence of the driving gap, the moving contact **9** will not contact the first push rod **150** when it rebounds, so the quick tripping device will not be driven. As shown in FIG. **40**, the huge electric repulsion force causes the moving contact **9** to be repulsed when the short-circuit current flows through the circuit breaker of the present invention, and a rotation angle of the moving contact **9** is much greater than a rebound amplitude that appears when the moving contact **9** and the static contact **18** are closed. Therefore, the moving

contact **9** will turn around the driving gap and then contact the first push rod driven end to drive the first push rod **150** to rotate, and the first push rod driving end drives the operating mechanism **100** to trip (even if the latch **13** releases the latching fit from the jump buckle **60**), so that the circuit breaker is quickly tripped or disconnected, avoiding the moving contact **9** and the static contact **18** from being closed again.

(173) It should be pointed out that the “the moving contact **9** is repelled by the short-circuit current” refers that: in the case that the short-circuit current flows through the closed moving contact **9** and U-shaped static contact **18**, a large electric repulsion force is generated between the moving contact **9** and the static contact **18** since the U-shaped static contact **18** has a short-circuit current in the opposite direction, so that the moving contact **9** and the static contact **18** are disconnected.

(174) Further, the first push rod driven end includes a driven protrusion or a driven groove.

(175) Further, the moving contact **9** includes a driving groove or a driving protrusion.

(176) Specifically, the first push rod driven end and the moving contact **9** may be cooperated by means of the driven protrusion and the driving protrusion, or by means of the driven groove and the driving protrusion, or by means of the driven groove and the driving groove, or by means of the driven protrusion and the driving groove.

(177) Further, as shown in FIGS. **39-40**, the moving contact **9** includes a moving conductive rod **90**, the moving conductive rod **90** including a contact protrusion **906** that is in driving fit with the first push rod **150**, a driving gap being formed between the contact protrusion **906** and the first push rod driven end. Further, as shown in FIG. **40**, the middle part of the first push rod **150** is pivotally disposed on the contact support **110**. The first push rod **150** includes a first push rod driven arm **150-1** (i.e., the first push rod driven end) and a first push rod driving arm **150-2** (i.e., the first push rod driving end) which are disposed at both ends, respectively and are in driving fit with the moving contact **9** and the intermediate transmission structure, respectively; and a driving gap is formed between the first push rod driven arm **150-1** and the contact support **906**. Further, as shown in FIG. **40**, the first push rod **150** further includes a first push rod mounting portion **150** pivotally disposed on the contact support **110**, wherein one end of the first push rod driven arm **150-1** and one end of the first push rod driving arm **150-2** are connected to the first push rod mounting portion **150**, respectively. Further, as shown in FIG. **39** and FIG. **40**, the first push rod **150** is pivotally disposed on the contact support **110** through the fourth spring shaft **202**.

(178) Further, as shown in FIGS. **39-42**, the quick tripping device further includes an intermediate transmission structure and a second push rod **180**, wherein the first push rod driving end is in driving fit with the second push rod **180** through the intermediate transmission structure, and the second push rod **180** is in driving fit with the operating mechanism **100** to drive the operating mechanism **100** to trip. Further, as shown in FIG. **41**, the second push rod **180** is in driving fit with the re-buckle **15** of the operating mechanism **100**.

(179) Further, as shown in FIGS. **39-42**, the intermediate transmission structure includes a first intermediate push rod **160**, a first intermediate shaft **161**, a second intermediate push rod **170-1** and a second intermediate shaft **170-2**, wherein the first intermediate push rod **160** is in driving fit with the first push rod driving end, the first intermediate shaft **161** is rotatably disposed around its axis line, and the first intermediate push rod **160** and the second intermediate push rod **170-1** are fixedly connected to the first intermediate shaft **161** respectively, such that the first intermediate push rod **160**, the first intermediate shaft **161** and the second intermediate shaft **170-2** rotate synchronously; one end of the second intermediate shaft **170-2** is connected to the second intermediate push rod **170-1**; and the other end of the second intermediate shaft **170-2** is in driving fit with the second push rod **180**. Further, as shown in FIGS. **39-42**, the first intermediate shaft **161** is inserted on the unit housing **120**, and an inner end and an outer end of the first intermediate shaft **161** are in driving connection to the first intermediate push rod **160** and the second intermediate push rod **170-1**, respectively. Further, as shown in FIG. **38**, the unit housing **120** is provided with an intermediate

shaft insertion hole **120-8** for the first intermediate shaft **161** to be inserted. It should be pointed out that the first intermediate shaft **161** may also be rotatably disposed on the bracket **50** of the operating mechanism **100**.

(180) Further, as shown in FIG. **42**, the first intermediate shaft **161** is provided with a shaft limiting plane **161-0** at one end, the second intermediate push rod **170** is provided with a second intermediate push rod hole **170**, and a sidewall of the second intermediate push rod hole **170** is provided with a hole limiting plane in limiting fit with the shaft limiting plane **161-0**.

(181) Further, as shown in FIGS. **40-42**, the first intermediate push rod **160** includes a first intermediate push rod driven arm **160-1** and a first intermediate push rod limiting arm **160-2** which are in driving fit with the first push rod **150**. As shown in FIGS. **39** and **40**, the quick tripping device further includes a push rod limiting protrusion **120-9** which is in limiting fit with the first intermediate push rod limiting arm **160-2**. Further, as shown in FIGS. **40-42**, the middle part of the first intermediate push rod **160** is fixedly connected to the first intermediate shaft **161**.

(182) Further, as shown in FIGS. **39** and **40**, the first push rod **150** and the push rod limiting protrusion **120-9** are located on both sides of the first intermediate push rod **160**, respectively. Further, as shown in FIGS. **40-42**, the push rod limiting protrusion **120-9** is disposed on the unit housing **120**. It should be pointed out that the setting position of the push rod limiting protrusion **120-9** is not limited to one of the above, as long as it can play the function of limiting the swing amplitude of the first intermediate push rod **160**.

(183) Further, as shown in FIG. **41**, the second push rod **180** is of a triangular plate-like structure, at one vertex angle of which a push rod driven hole **180-2** is formed for the second intermediate shaft **170-2** to be inserted to be in driving fit with the push rod driven hole, at a second vertex angle of which a second push rod shaft **4** is disposed pivotally, and at a third vertex angle of which a push rod driving finger **180-1** that is in driving fit with the re-buckle **15** is disposed. Further, as shown in FIG. **41**, the second push rod **180** is pivotally disposed outside the unit housing **120** through the second push rod shaft **4**.

(184) Further, the re-buckle **15** includes a re-buckle driven column **15-9** in driving fit with the second push rod **180**, and the re-buckle driven column **15-9** is in driving fit with the push rod driving finger **180-1**.

(185) Further, as shown in FIG. **41**, the first intermediate shaft **161** is inserted on the unit housing **120**, and two ends of the first intermediate shaft **161** are located inside and outside the unit housing **120** respectively. The first push rod **150** and the first intermediate push rod **160** are respectively disposed inside the unit housing **120**, and the second intermediate push rod **170-1**, the second intermediate shaft **170-2** and the second push rod **180** are respectively disposed outside the unit housing **120**.

(186) Further, as shown in FIG. **44**, the circuit breaker of the present invention includes a plurality of circuit breaker poles **300** arranged side by side, wherein each circuit breaker pole including an independent first push rod **150**, a first intermediate push rod **160**, a first intermediate shaft **161**, a second intermediate push rod **170-1** and a second intermediate shaft **170-2**. Further, as shown in FIG. **44**, each of the circuit breaker poles includes an independent second push rod **180**; or two adjacent circuit breaker poles share one second push rod **180**.

(187) Specifically, as shown in FIG. **44**, the circuit breaker of the present invention includes three circuit breaker poles **300** arranged side by side, wherein the circuit breaker poles **300** on the left side and in the middle share one second push rod **180**, and the circuit breaker pole **300** on the right side includes an independent second push rod **180**.

(188) Further, as shown in FIGS. **38** and **43**, each of the unit housing **120** includes a first connecting lug **120-1** and a second connecting lug **120-3** which are disposed on the sidewall at one end. The second push rod shaft **4** passes through each second connecting lug **120-3** to connect the unit housing **120** together. The circuit breaker further includes a second connecting shaft **4a**, wherein the second connecting shaft **4a** passes through the bracket **50** of the operating mechanism

100, and each first connecting lug **120-1** connects the operating mechanism **100** and the unit housing **120** together.

(189) The above content is a further detailed description of the present invention in conjunction with specific preferred embodiments, and it cannot be considered that the specific embodiments of the present invention are limited to these descriptions. For those of ordinary skill in the technical field to which the present invention belongs, several simple deductions or substitutions can be made without departing from the concept of the present invention, which should be regarded as falling within the protection scope of the present invention.

Claims

1. An operating mechanism of a circuit breaker, comprising a bracket, a rocker arm assembly and a jump buckle pivotally disposed on the bracket respectively, as well as a first crank, a first spring, a first connecting rod and a contact support, one end of the first crank is pivotally disposed on the jump buckle around a first axis, and another end of the first crank is rotatably connected to one end of the first connecting rod; one end of the first spring is connected to the rocker arm assembly, and another end of the first spring is connected to a rotating connection position of the first crank and the first connecting rod; wherein the operating mechanism further comprises slide rails, a slider and a second connecting rod; the slider is slidably disposed on the slide rails and is rotatably connected to another end of the first connecting rod; one end of the second connecting rod is rotatably connected to the slider, and another end of the second connecting rod is rotatably connected to the contact support; and the contact support is driven to rotate around a third axis.
2. The operating mechanism of the circuit breaker as claimed in claim 1, wherein the operating mechanism further comprises an auxiliary limiting structure, one end of the auxiliary limiting structure is rotatably connected to the second connecting rod, and another end of the auxiliary limiting structure is rotatably connected to the bracket or a housing of the circuit breaker.
3. The operating mechanism of the circuit breaker as claimed in claim 2, wherein the auxiliary limiting structure is a third crank, one end of the third crank is rotatably connected to the second connecting rod, and another end of the third crank is rotatably connected to the bracket or the housing of the circuit breaker.
4. The operating mechanism of the circuit breaker as claimed in claim 1, wherein the bracket comprises two bracket arms disposed to face each other at intervals, and each bracket arm is provided with a slide rail, and two ends of the slider are respectively slidably disposed on the two slide rails.
5. The operating mechanism of the circuit breaker as claimed in claim 4, wherein the slide rails are slide holes, and the slider is a sliding shaft, two ends of which are respectively disposed in the two slide holes.
6. The operating mechanism of the circuit breaker as claimed in claim 1, wherein end of the second connecting rod is rotatably connected to the contact support through a first connecting shaft; and the bracket comprises a second avoidance hole for the first connecting shaft to pass through for avoiding the first connecting shaft.
7. The operating mechanism of the circuit breaker as claimed in claim 1, wherein the slide rails are disposed on the bracket or a housing of the circuit breaker, and the slide rails are of a groove-like structure or a hole-like structure.
8. The operating mechanism of the circuit breaker as claimed in claim 1, wherein the operating mechanism further comprises a latch and a re-buckle which are pivotally disposed on the bracket respectively; the jump buckle is in latching fit with the latch, and the latch is in limiting fit with the re-buckle; the rocker arm assembly comprises a handle, a rocker arm fixedly connected to the handle, and a reset structure for driving the jump buckle to rotate to re-buckle with the latch, the rocker arm being pivotally disposed on the bracket; the first crank comprises a crank limiting

portion, which is in limiting fit with the jump buckle; two ends of a swing stroke of the rocker arm are a first stroke end and a second stroke end, respectively; two ends of the first spring are a first spring end and a second spring end, respectively, which are respectively connected to the rocker arm assembly and the first crank.

9. The operating mechanism of the circuit breaker as claimed in claim 8, wherein in a case that the operating mechanism is in a closed state, the rocker arm swings toward the second stroke end and drives the first spring end to rotate around the second spring end, until the first spring passes a first dead center position; the first spring drives the first crank to rotate in a second direction and drives the rocker arm to swing to the second stroke end; the first crank drives the slider through the first connecting rod to slide along the slide rails; and the slider drives the contact support through the second connecting rod to rotate in a first direction to a breaking position, so that the operating mechanism is switched to a disconnected state; in a case that the operating mechanism is in the disconnected state, the rocker arm swings toward the first stroke end and drives the first spring end to rotate around the second spring end, until the first spring passes the first dead center position; the first spring drives the first crank to rotate in the first direction, such that the crank limiting portion is in limiting fit with the jump buckle, thereby preventing the first crank from rotating in the first direction; meanwhile, the first spring drives the rocker arm to swing to the first stroke end, and the first crank drives the slider through the first connecting rod to slide along the slide rails; and the slider drives the contact support through the second connecting rod to rotate in the first direction to a closing position, so that the operating mechanism is switched to a closed state; the first direction and the second direction are opposite to each other, in the case that the operating mechanism is in the closed state, the re-buckle rotates to release the limiting fit from the latch, and the latch rotates to release the latching fit from the jump buckle; the jump buckle rotates and drives the first crank to rotate synchronously, the first crank drives the slider through the first connecting rod to slide along the slide rails, and meanwhile, the slider drives the contact support through the second connecting rod to rotate in the second direction to the breaking position; the first spring drives the rocker arm to swing toward the second stroke end to the reset structure till being in limiting fit with the jump buckle, and the operating mechanism is switched to a tripped state; and in a case that the operating mechanism is in the tripped state, the rocker arm swings to the second stroke end, and the rocker arm drives the jump buckle through the reset structure to rotate to be in latching fit with the latch; and meanwhile, the latch rotates to be in limiting fit with the re-buckle, and the operating mechanism is switched to the disconnected state.

10. A circuit breaker, wherein comprising the operating mechanism of the circuit breaker as claimed in claim 1.

11. An operating mechanism of a circuit breaker, comprising a bracket; a rocker arm assembly and a jump buckle pivotally disposed on the bracket respectively, as well as a first crank, a first spring, a first connecting rod and a contact support, one end of the first crank is pivotally disposed on the jump buckle around a first axis, and another end of the first crank is rotatably connected to one end of the first connecting rod; one end of the first spring is connected to the rocker arm assembly, and another end of the first spring is connected to a rotating connection position of the first crank and the first connecting rod; wherein the operating mechanism further comprises slide rails, a slider, a second crank and a second connecting rod; the slider is slidably disposed on the slide rails and is rotatably connected to another end of the first connecting rod; the second crank comprises a second crank supporting portion, a second crank connecting portion, and a second crank driving portion, wherein the second crank is pivotally disposed through the second crank supporting portion, and the second crank is connected to the contact support through the second crank driving portion; one end of the second connecting rod is rotatably connected to the slider, and another end of the second connecting rod is rotatably connected to the second crank connecting portion; and the contact support is driven to rotate around a third axis.

12. The operating mechanism of the circuit breaker as claimed in claim 11, wherein the second

crank is pivotally disposed on the bracket or a housing of the circuit breaker through the second crank supporting portion; the second crank supporting portion and the second crank driving portion are respectively disposed on two ends of the second crank; and the second crank connecting portion is disposed between the second crank supporting portion and the second crank driving portion.

13. The operating mechanism of the circuit breaker as claimed in claim 12, wherein the second crank driving portion is rotatably connected to the contact support through a linkage shaft.

14. The operating mechanism of the circuit breaker as claimed in claim 13, wherein the housing of the circuit breaker is provided with a first avoidance hole for the linkage shaft to pass through for avoiding the linkage shaft; end of the second connecting rod is rotatably connected to the second crank connecting portion through a first connecting shaft; and the bracket is provided with a second avoidance hole for the first connecting shaft to pass through for avoiding the first connecting shaft.

15. The operating mechanism of the circuit breaker as claimed in claim 11, wherein a center of rotation of the second crank is a ninth axis, and the ninth axis is parallel to or coincided with the third axis.
